

Power Analysis for Confirmatory Factor Analysis: A Practical Tutorial

Analytical and Simulation-Based Approaches

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This tutorial presents a pedagogical introduction to statistical power analysis in Confirmatory Factor Analysis (CFA). Proper sample size planning requires assessing the probability of detecting overall model misfit, either through analytical methods or simulation-based approaches. Standard analytical tools rely on asymptotic normal theory, which can break down under non-normality or missing data. This tutorial demonstrates both strategies — analytical (using `semPower` with `lavaan` syntax) and Monte Carlo simulation (using `simsem`) — side by side, using the WHOQOL-BREF four-factor model as a unifying case study. We illustrate how simulation complements analytical calculations by enabling researchers to evaluate parameter-level power, coverage, and relative bias under realistic scenarios involving model misspecification, missing data, and non-normal continuous indicators. We argue that analytical methods provide a fast and theoretically grounded starting point for sample size planning, while simulation-based approaches allow those conclusions to be stress-tested against the specific population structure, misspecification form, and data conditions of the study at hand.

1 Introduction

Statistical power in CFA is the probability of correctly rejecting a misspecified model — or of detecting a parameter of interest when it is truly non-zero in the population (MacCallum et al., 1996). Despite its importance, power is rarely reported in applied CFA work: researchers frequently fall back on arbitrary rules of thumb linking required sample size to the number of observed variables or estimated parameters, rather than conducting formal prospective calculations (Aguirre-Urreta et al., 2024; Wolf et al., 2013).

Two main paradigms address power analysis in CFA:

1. **Analytical methods:** Rely on the non-central χ^2 distribution to determine the sample size required for detecting global or local model misfit, typically quantified through the RMSEA (Jak et al., 2021; MacCallum et al., 1996).
2. **Simulation-based methods:** Use Monte Carlo data generation to estimate power, bias, and coverage for specific parameters or overall fit indices across varying sample sizes, without relying on the asymptotic assumptions that constrain analytical solutions (Muthén & Muthén, 2002; Pornprasertmanit et al., 2021).

This document provides a comprehensive tutorial covering both paradigms: analytical power analysis using the `semPower` package, and Monte Carlo simulation using the `simsem` package (Moshagen & Bader, 2024; Pornprasertmanit et al., 2021).

Reproducibility Note

For local development and computational convenience, simulations in this tutorial use 1,500 replications. Production-quality applied research should increase this to 5,000 or more. Results throughout this tutorial therefore carry more sampling variability than would be expected in a final study. All analytical and simulation parameters are centrally defined in `helpers.R` and `models.R` in `Scripts/DataAppendixScripts`.

2 The Running Example: WHOQOL-BREF Population CFA Model

All power analyses in this tutorial are anchored in the World Health Organization Quality of Life-BREF (WHOQOL-BREF) four-factor measurement model. Meaningful power analysis requires an explicit population model — a formal representation of the “true” covariance structure from which data are presumed to be generated (Feng & Hancock, 2023). By grounding this model in meta-analytic evidence rather than in arbitrary parameter choices, we ensure that sample size targets reflect realistic measurement conditions (Lin & Yao, 2022; Mosqueira Taipe et al., 2026).

We specify four population variants to support sensitivity analyses:

1. **Meta-analytic** (`popmodel`): Loadings and residual variances matching the pooled estimates from 1.
2. **Naïve** (`naivemodel`): A pessimistic baseline derived from the lowest reported reliability values per domain (Mosqueira Taipe et al., 2026).
3. **Optimistic** (`optmodel`): A best-case scenario derived from the highest reported reliability values per domain (Mosqueira Taipe et al., 2026).
4. **Misspecification** (`h1model`): A cross-loading model representing the alternative hypothesis H_1 , with three additional cross-loadings based on the meta-EFA results of 1.

2.1 Population model

The primary population model (`popmodel`) operationalizes a meta-analytic scenario based on 1, whose pooled EFA estimates across 16 international studies provide the factor loadings (λ). Key features of this model are:

- **Interfactor correlations (ϕ):** Fixed at 0.20, a conservative midpoint of the reported range 0.08–0.31 (Lin & Yao, 2022).
- **Residual covariance:** Items $Q3$ and $Q4$ share a residual covariance ($r_{Q3,Q4|\text{physical}} = 0.30$), preserving the subfactor signal identified in the five-factor EFA solution (Lin & Yao, 2022).
- **Factor structure:** Item $Q8$ is retained in the Environmental domain despite loading more strongly on the Psychological domain in the meta-EFA. This constitutes the null hypothesis (H_0) version of the population, strictly maintaining the original four-factor structure without cross-loadings.

The `lavaan` syntax for this model is shown below; Figure ?? provides a visual representation.

```
# Factor loadings: empirical estimates from Lin & Yao (2022)
psycho      =~ 0.92*Q5 + 0.81*Q6 + 0.94*Q7 +
              0.73*Q11 + 0.75*Q19 + 0.63*Q26

physical    =~ 0.84*Q3 + 0.83*Q4 + 0.63*Q10 +
              0.71*Q15 + 0.68*Q16 + 0.89*Q17 + 0.61*Q18

social      =~ 0.94*Q20 + 0.89*Q21 + 0.78*Q22

environment =~ 0.37*Q8 + 0.49*Q9 + 0.79*Q12 + 0.81*Q13 +
              0.77*Q14 + 0.75*Q23 + 0.64*Q24 + 0.73*Q25

# Residual covariance Q3~~Q4
# r(Q3,Q4|physical) = 0.30 (method bias / conceptual proximity)
# cov = 0.30 × sqrt(1-0.84²) × sqrt(1-0.83²) = 0.0908
Q4 ~~ 0.0908*Q3

# Factor variances (=1) and interfactor correlations
# Range reported in Lin & Yao (2022): 0.08 to 0.31
# Conservative midpoint = 0.20 for all pairs
psycho      ~~ 1*psycho      + 0.2*physical + 0.2*social + 0.2*environment
physical    ~~ 1*physical    + 0.2*social   + 0.2*environment
social      ~~ 1*social      + 0.2*environment
environment ~~ 1*environment
```

```

# Residual variances: = 1 2 (standardised indicators)
Q5  ~~ 0.1536*Q5    # 1  0.922
Q6  ~~ 0.3439*Q6    # 1  0.812
Q7  ~~ 0.1164*Q7    # 1  0.942
Q11 ~~ 0.4671*Q11   # 1  0.732
Q19 ~~ 0.4375*Q19   # 1  0.752
Q26 ~~ 0.6031*Q26   # 1  0.632
Q3   ~~ 0.2944*Q3    # 1  0.842
Q4   ~~ 0.3111*Q4    # 1  0.832
Q10  ~~ 0.6031*Q10   # 1  0.632
Q15  ~~ 0.4959*Q15   # 1  0.712
Q16  ~~ 0.5376*Q16   # 1  0.682
Q17  ~~ 0.2079*Q17   # 1  0.892
Q18  ~~ 0.6279*Q18   # 1  0.612
Q20  ~~ 0.1164*Q20   # 1  0.942
Q21  ~~ 0.2079*Q21   # 1  0.892
Q22  ~~ 0.3916*Q22   # 1  0.782
Q8   ~~ 0.8631*Q8    # 1  0.372 - lowest loading; bridge item (betweenness=40.5)
Q9   ~~ 0.7599*Q9    # 1  0.492
Q12  ~~ 0.3759*Q12   # 1  0.792
Q13  ~~ 0.3439*Q13   # 1  0.812
Q14  ~~ 0.4071*Q14   # 1  0.772
Q23  ~~ 0.4375*Q23   # 1  0.752
Q24  ~~ 0.5904*Q24   # 1  0.642
Q25  ~~ 0.4671*Q25   # 1  0.732

```

Source: [Population Models](#)

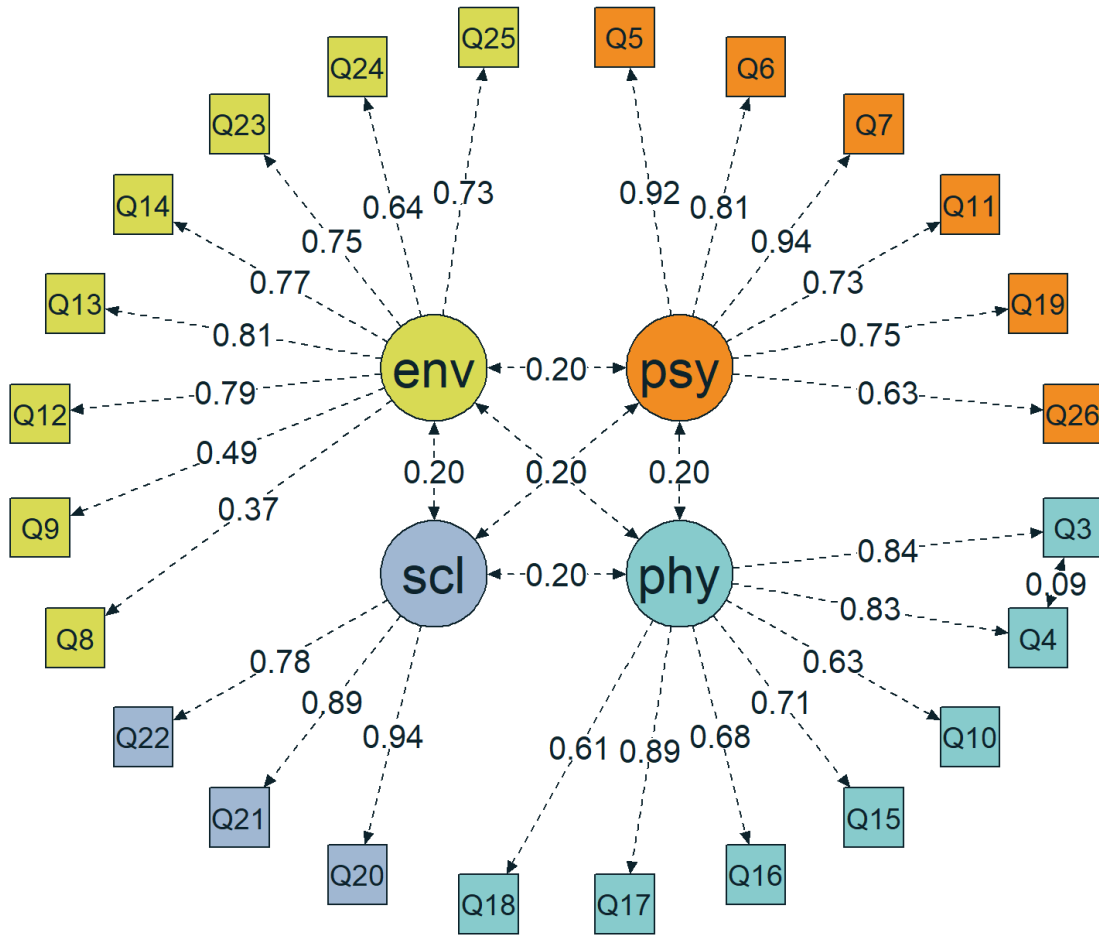


Figure 1: Path diagram of the popmodel

Source: [Population Models](#)

2.2 Naive model

The naive model provides an empirically grounded pessimistic scenario. Factor loadings are derived via the Spearman-Brown prediction formula from the lowest Cronbach's α values identified in the literature (Mosqueira Taípe et al., 2026), yielding conservative parameter estimates that push sample size requirements upward:

- **Approach:** Assuming equal loadings within each domain, $\lambda = \sqrt{\alpha/(k - \alpha(k - 1))}$ and $\theta = 1 - \lambda^2$.
- **Alpha sources:** Lowest domain alphas from a single study identified in the systematic review by Mosqueira Taípe et al. (2026) (Physical $\alpha = 0.672$; Psychological $\alpha = 0.704$; Social $\alpha = 0.630$; Environmental $\alpha = 0.709$), all within the $M - 1SD$ range reported by 1.
- **Interfactor correlations (ϕ):** Fixed at 0.30 (the upper bound from 1), making factors harder to distinguish and yielding a deliberately conservative power estimate.
- **Derived parameters:** Physical ($\lambda = 0.4758$, $\theta = 0.7736$); Psychological ($\lambda = 0.5328$, $\theta = 0.7161$); Social ($\lambda = 0.6017$, $\theta = 0.6379$); Environmental ($\lambda = 0.4832$, $\theta = 0.7665$). The residual covariance between $Q3$ and $Q4$ is adjusted to 0.2321.

```
# Factor loadings: tau-equivalent (uniform per domain)
psycho      =~ 0.5328*Q5  + 0.5328*Q6  + 0.5328*Q7  +
              0.5328*Q11 + 0.5328*Q19 + 0.5328*Q26

physical    =~ 0.4758*Q3  + 0.4758*Q4  + 0.4758*Q10 +
              0.4758*Q15 + 0.4758*Q16 + 0.4758*Q17  + 0.4758*Q18

social      =~ 0.6017*Q20 + 0.6017*Q21 + 0.6017*Q22

environment =~ 0.4832*Q8  + 0.4832*Q9  + 0.4832*Q12 + 0.4832*Q13 +
              0.4832*Q14 + 0.4832*Q23  + 0.4832*Q24 + 0.4832*Q25

# Residual covariance Q3~~Q4
# r(Q3,Q4|physical) = 0.30; cov = 0.30 × (1 - 0.47582) = 0.2321
Q4 ~~ 0.2321*Q3

# Factor variances (=1) and interfactor correlations
# = 0.30: upper bound of Lin & Yao (2022) range - pessimistic scenario
psycho      ~~ 1*psycho      + 0.3*physical + 0.3*social + 0.3*environment
physical    ~~ 1*physical    + 0.3*social   + 0.3*environment
social      ~~ 1*social      + 0.3*environment
environment ~~ 1*environment

# Residual variances: = 1 - 2 per domain
Q5  ~~ 0.7161*Q5    # Psychological: = 1 - 0.53282 = 0.7161
Q6  ~~ 0.7161*Q6
Q7  ~~ 0.7161*Q7
Q11 ~~ 0.7161*Q11
Q19 ~~ 0.7161*Q19
```

Q26 ~~ 0.7161*Q26
 Q3 ~~ 0.7736*Q3 # Physical: = 1 $0.4758^2 = 0.7736$
 Q4 ~~ 0.7736*Q4
 Q10 ~~ 0.7736*Q10
 Q15 ~~ 0.7736*Q15
 Q16 ~~ 0.7736*Q16
 Q17 ~~ 0.7736*Q17
 Q18 ~~ 0.7736*Q18
 Q20 ~~ 0.6379*Q20 # Social: = 1 $0.6017^2 = 0.6379$
 Q21 ~~ 0.6379*Q21
 Q22 ~~ 0.6379*Q22
 Q8 ~~ 0.7665*Q8 # Environmental: = 1 $0.4832^2 = 0.7665$
 Q9 ~~ 0.7665*Q9
 Q12 ~~ 0.7665*Q12
 Q13 ~~ 0.7665*Q13
 Q14 ~~ 0.7665*Q14
 Q23 ~~ 0.7665*Q23
 Q24 ~~ 0.7665*Q24
 Q25 ~~ 0.7665*Q25

Source: [Population Models](#)

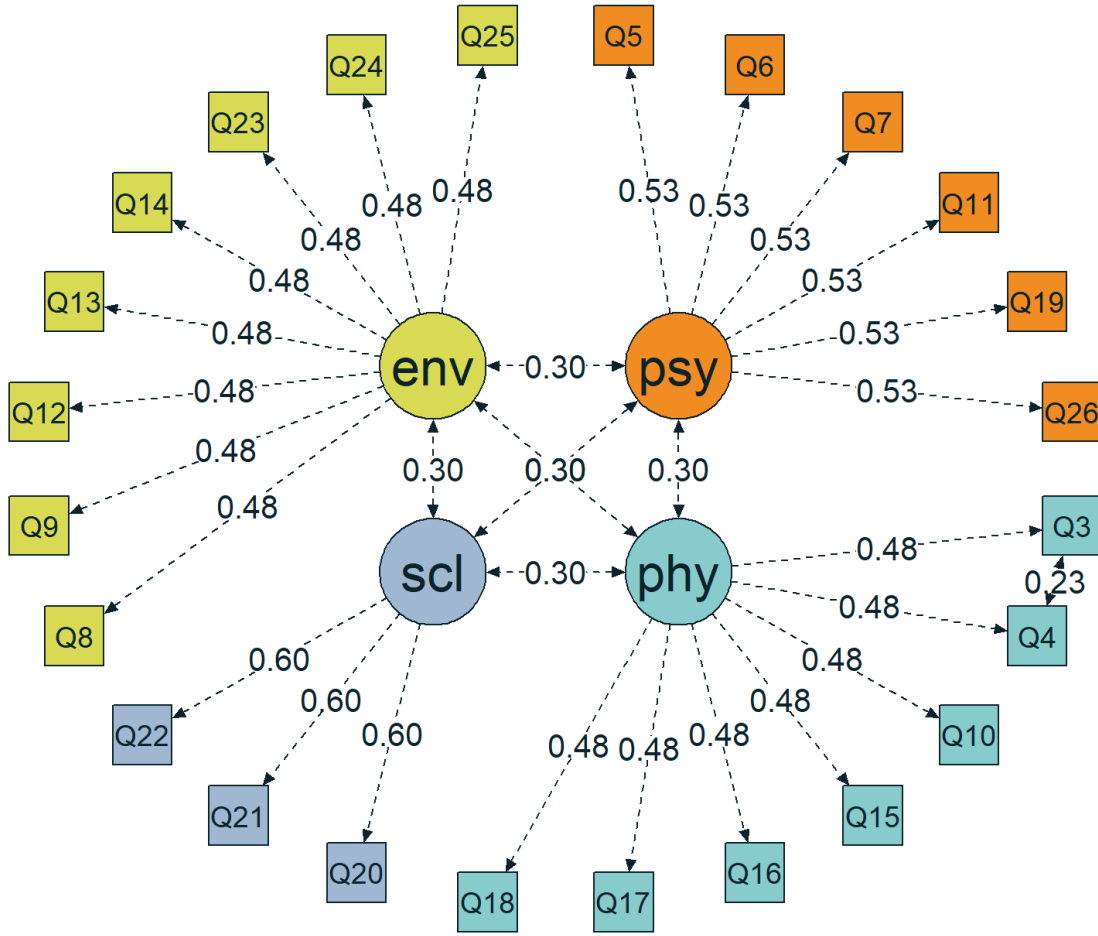


Figure 2: Path diagram of the naive model

Source: [Population Models](#)

2.3 Optimistic model

The optimistic model applies the same Spearman-Brown derivation but draws on the highest observed Cronbach's α values per domain, producing stronger loadings and lower residual variances relative to the naive model (Mosqueira Taipe et al., 2026):

- **Alpha sources:** Maximum domain alphas from the literature (Mosqueira Taipe et al., 2026) (Physical $\alpha = 0.860$; Psychological $\alpha = 0.860$; Social $\alpha = 0.810$; Environmental

$\alpha = 0.849$), all confirmed within the $M + 1SD$ range from

1.

- **Interfactor correlations (ϕ):** Fixed at 0.08 (the lower bound from 1), so that factors are well distinguished and sample size requirements are reduced.
- **Derived parameters:** Physical ($\lambda = 0.6837$, $\theta = 0.5326$); Psychological ($\lambda = 0.7113$, $\theta = 0.4941$); Social ($\lambda = 0.7661$, $\theta = 0.4130$); Environmental ($\lambda = 0.6424$, $\theta = 0.5873$). The residual covariance between $Q3$ and $Q4$ is adjusted to 0.1598.

```
# Factor loadings: tau-equivalent (uniform per domain)
psycho      =~ 0.7113*Q5  + 0.7113*Q6  + 0.7113*Q7  +
              0.7113*Q11 + 0.7113*Q19 + 0.7113*Q26

physical     =~ 0.6837*Q3  + 0.6837*Q4  + 0.6837*Q10 +
              0.6837*Q15 + 0.6837*Q16 + 0.6837*Q17  + 0.6837*Q18

social       =~ 0.7661*Q20 + 0.7661*Q21 + 0.7661*Q22

environment  =~ 0.6424*Q8  + 0.6424*Q9  + 0.6424*Q12 + 0.6424*Q13 +
              0.6424*Q14 + 0.6424*Q23  + 0.6424*Q24 + 0.6424*Q25

# Residual covariance Q3~~Q4
# r(Q3,Q4|physical) = 0.30; cov = 0.30 × (1 - 0.68372) = 0.1598
Q4 ~~ 0.1598*Q3

# Factor variances (=1) and interfactor correlations
# = 0.08: lower bound of Lin & Yao (2022) range - optimistic scenario
psycho      ~~ 1*psycho      + 0.08*physical + 0.08*social + 0.08*environment
physical     ~~ 1*physical    + 0.08*social  + 0.08*environment
social       ~~ 1*social      + 0.08*environment
environment  ~~ 1*environment

# Residual variances: = 1 - 2 per domain
Q5 ~~ 0.4941*Q5      # Psychological: = 1 - 0.71132 = 0.4941
Q6 ~~ 0.4941*Q6
Q7 ~~ 0.4941*Q7
Q11 ~~ 0.4941*Q11
Q19 ~~ 0.4941*Q19
Q26 ~~ 0.4941*Q26
Q3  ~~ 0.5326*Q3     # Physical: = 1 - 0.68372 = 0.5326
Q4  ~~ 0.5326*Q4
```

```

Q10 ~~ 0.5326*Q10
Q15 ~~ 0.5326*Q15
Q16 ~~ 0.5326*Q16
Q17 ~~ 0.5326*Q17
Q18 ~~ 0.5326*Q18
Q20 ~~ 0.4130*Q20   # Social:   = 1    $0.7661^2 = 0.4130$ 
Q21 ~~ 0.4130*Q21
Q22 ~~ 0.4130*Q22
Q8  ~~ 0.5873*Q8    # Environmental: = 1    $0.6424^2 = 0.5873$ 
Q9  ~~ 0.5873*Q9
Q12 ~~ 0.5873*Q12
Q13 ~~ 0.5873*Q13
Q14 ~~ 0.5873*Q14
Q23 ~~ 0.5873*Q23
Q24 ~~ 0.5873*Q24
Q25 ~~ 0.5873*Q25

```

Source: [Population Models](#)

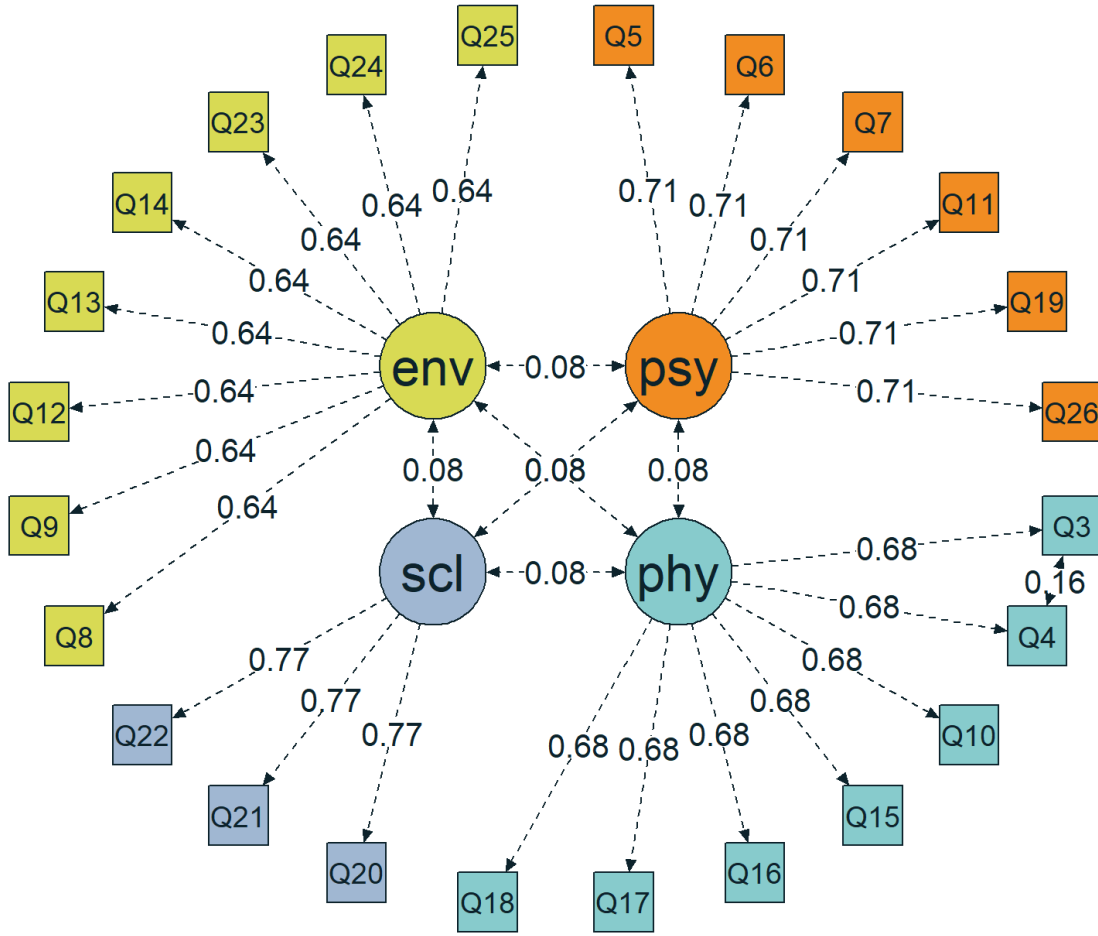


Figure 3: Path diagram of the optmodel

Source: [Population Models](#)

2.4 Misspecification model

This model defines the alternative hypothesis (H_1): a cross-loading variant of the four-factor structure in which three empirically motivated cross-loadings are added based on the meta-EFA results of 1 and the systematic review evidence in Mosqueira Taípe et al. (2026):

- **Cross-loading 1:** Item Q_8 on the Psychological factor ($\lambda = 0.55$). This item showed the highest betweenness centrality (40.50) and was assigned to the Psychological factor

in 9 of 16 studies (Lin & Yao, 2022).

- **Cross-loading 2:** Item *Q9* on the Psychological factor ($\lambda = 0.21$), identified as a secondary loading in the five-factor EFA solution.
- **Cross-loading 3:** Item *Q15* on the Environmental factor ($\lambda = 0.24$). “Mobility” conceptually bridges physical ability and environmental access, and this item showed the second-highest betweenness centrality (36.50) (Lin & Yao, 2022).

Residual variances for cross-loaded items were recalculated to preserve total item variance: $\theta = 1 - \lambda_{\text{primary}}^2 - \lambda_{\text{secondary}}^2 - 2\lambda_{\text{primary}}\lambda_{\text{secondary}}\phi$, where $\phi = 0.20$.

```
# Factor loadings (with three cross-loadings)
psycho      =~ 0.92*Q5 + 0.81*Q6 + 0.94*Q7 +
              0.73*Q11 + 0.75*Q19 + 0.63*Q26 +
              0.55*Q8 +                                     # cross-load (1)
              0.21*Q9                                     # cross-load (2)

physical    =~ 0.84*Q3 + 0.83*Q4 + 0.63*Q10 +
              0.71*Q15 + 0.68*Q16 + 0.89*Q17 + 0.61*Q18

social      =~ 0.94*Q20 + 0.89*Q21 + 0.78*Q22

environment =~ 0.37*Q8 + 0.49*Q9 + 0.79*Q12 + 0.81*Q13 +
              0.77*Q14 + 0.75*Q23 + 0.64*Q24 + 0.73*Q25 +
              0.24*Q15                                     # cross-load (3)

# Residual covariance Q3~~Q4 (fixed in both P2 and analyzeModel)
Q4 ~~ 0.0908*Q3

# Factor variances (=1) and interfactor correlations
psycho      ~~ 1*psycho + 0.2*physical + 0.2*social + 0.2*environment
physical    ~~ 1*physical + 0.2*social + 0.2*environment
social      ~~ 1*social + 0.2*environment
environment ~~ 1*environment

# Residual variances
Q5 ~~ 0.1536*Q5
Q6 ~~ 0.3439*Q6
Q7 ~~ 0.1164*Q7
Q11 ~~ 0.4671*Q11
Q19 ~~ 0.4375*Q19
Q26 ~~ 0.6031*Q26
Q3  ~~ 0.2944*Q3
```

```

Q4  ~~ 0.3111*Q4
Q10 ~~ 0.6031*Q10
Q15 ~~ 0.3701*Q15  # UPDATED: was 0.4959 - cross-load on environment
Q16 ~~ 0.5376*Q16
Q17 ~~ 0.2079*Q17
Q18 ~~ 0.6279*Q18
Q20 ~~ 0.1164*Q20
Q21 ~~ 0.2079*Q21
Q22 ~~ 0.3916*Q22
Q8   ~~ 0.4792*Q8   # UPDATED: was 0.8631 - cross-load on psycho
Q9   ~~ 0.6746*Q9   # UPDATED: was 0.7599 - cross-load on psycho
Q12  ~~ 0.3759*Q12
Q13  ~~ 0.3439*Q13
Q14  ~~ 0.4071*Q14
Q23  ~~ 0.4375*Q23
Q24  ~~ 0.5904*Q24
Q25  ~~ 0.4671*Q25

```

Source: [Population Models](#)

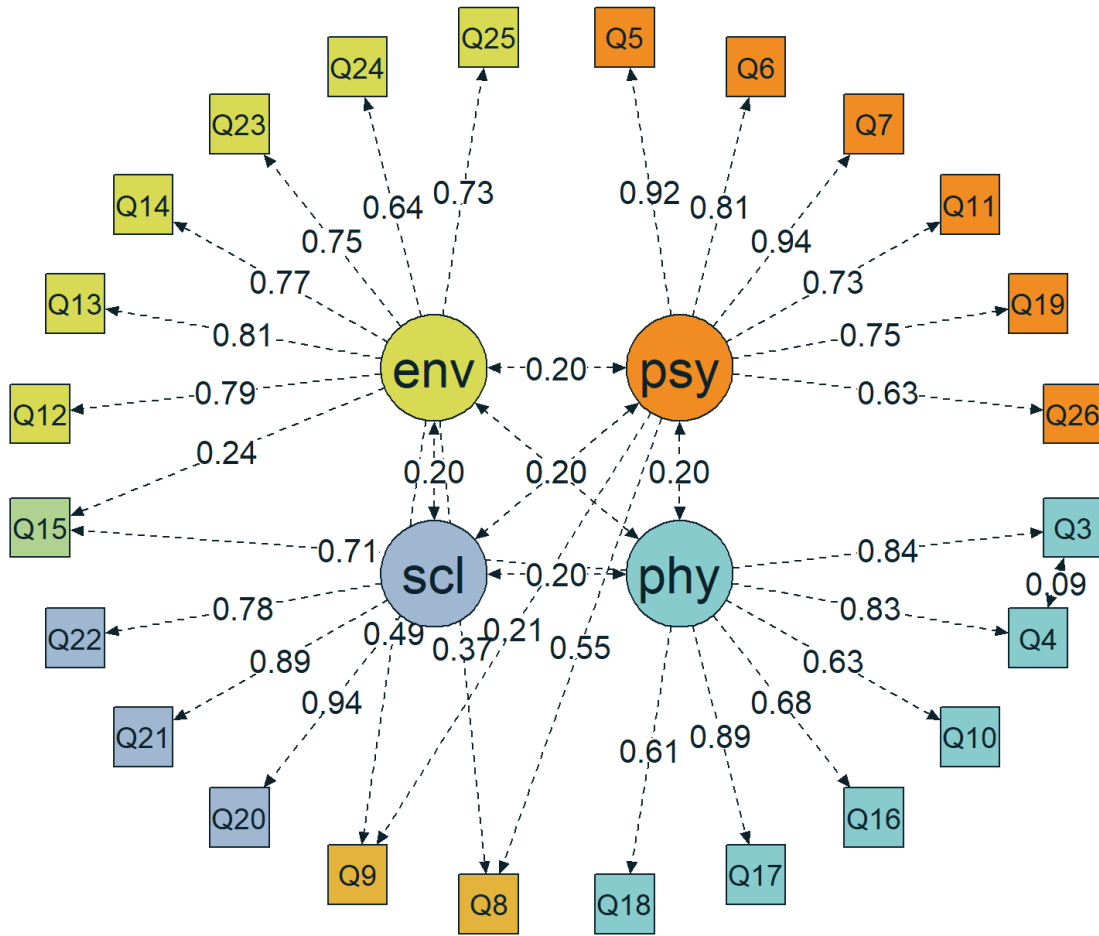


Figure 4: Path diagram of the h1model

Source: [Population Models](#)

2.5 Summary of population models

Table ?? presents all parameters of the four population models. Table ?? summarizes key features across models — minimum and maximum factor loadings, mean loading, modal interfactor correlation, minimum and maximum residual variances, and the residual correlation between $Q3$ and $Q4$ — to facilitate direct comparison.

Table 1: Summary of population models used for sensitivity analysis

Table 1: CFA Population Models Parameter

parameters	popmodel	naivemodel	optmodel	h1model
psycho=~Q5	0.9200	0.5328	0.7113	0.9200
psycho=~Q6	0.8100	0.5328	0.7113	0.8100
psycho=~Q7	0.9400	0.5328	0.7113	0.9400
psycho=~Q11	0.7300	0.5328	0.7113	0.7300
psycho=~Q19	0.7500	0.5328	0.7113	0.7500
psycho=~Q26	0.6300	0.5328	0.7113	0.6300
physical=~Q3	0.8400	0.4758	0.6837	0.8400
physical=~Q4	0.8300	0.4758	0.6837	0.8300
physical=~Q10	0.6300	0.4758	0.6837	0.6300
physical=~Q15	0.7100	0.4758	0.6837	0.7100
physical=~Q16	0.6800	0.4758	0.6837	0.6800
physical=~Q17	0.8900	0.4758	0.6837	0.8900
physical=~Q18	0.6100	0.4758	0.6837	0.6100
social=~Q20	0.9400	0.6017	0.7661	0.9400
social=~Q21	0.8900	0.6017	0.7661	0.8900
social=~Q22	0.7800	0.6017	0.7661	0.7800
environment=~Q8	0.3700	0.4832	0.6424	0.3700
environment=~Q9	0.4900	0.4832	0.6424	0.4900
environment=~Q12	0.7900	0.4832	0.6424	0.7900
environment=~Q13	0.8100	0.4832	0.6424	0.8100
environment=~Q14	0.7700	0.4832	0.6424	0.7700
environment=~Q23	0.7500	0.4832	0.6424	0.7500
environment=~Q24	0.6400	0.4832	0.6424	0.6400
environment=~Q25	0.7300	0.4832	0.6424	0.7300
Q3~~Q4	0.0908	0.2321	0.1598	0.0908
psycho~~psycho	1.0000	1.0000	1.0000	1.0000
psycho~~physical	0.2000	0.3000	0.0800	0.2000
psycho~~social	0.2000	0.3000	0.0800	0.2000
psycho~~environment	0.2000	0.3000	0.0800	0.2000
physical~~physical	1.0000	1.0000	1.0000	1.0000
physical~~social	0.2000	0.3000	0.0800	0.2000
physical~~environment	0.2000	0.3000	0.0800	0.2000
social~~social	1.0000	1.0000	1.0000	1.0000
social~~environment	0.2000	0.3000	0.0800	0.2000
environment~~environment	1.0000	1.0000	1.0000	1.0000
Q5~~Q5	0.1536	0.7161	0.4941	0.1536
Q6~~Q6	0.3439	0.7161	0.4941	0.3439

parameters	popmodel	naivemodel	optmodel	h1model
Q7~~Q7	0.1164	0.7161	0.4941	0.1164
Q11~~Q11	0.4671	0.7161	0.4941	0.4671
Q19~~Q19	0.4375	0.7161	0.4941	0.4375
Q26~~Q26	0.6031	0.7161	0.4941	0.6031
Q3~~Q3	0.2944	0.7736	0.5326	0.2944
Q4~~Q4	0.3111	0.7736	0.5326	0.3111
Q10~~Q10	0.6031	0.7736	0.5326	0.6031
Q15~~Q15	0.4959	0.7736	0.5326	0.3701
Q16~~Q16	0.5376	0.7736	0.5326	0.5376
Q17~~Q17	0.2079	0.7736	0.5326	0.2079
Q18~~Q18	0.6279	0.7736	0.5326	0.6279
Q20~~Q20	0.1164	0.6379	0.4130	0.1164
Q21~~Q21	0.2079	0.6379	0.4130	0.2079
Q22~~Q22	0.3916	0.6379	0.4130	0.3916
Q8~~Q8	0.8631	0.7665	0.5873	0.4792
Q9~~Q9	0.7599	0.7665	0.5873	0.6746
Q12~~Q12	0.3759	0.7665	0.5873	0.3759
Q13~~Q13	0.3439	0.7665	0.5873	0.3439
Q14~~Q14	0.4071	0.7665	0.5873	0.4071
Q23~~Q23	0.4375	0.7665	0.5873	0.4375
Q24~~Q24	0.5904	0.7665	0.5873	0.5904
Q25~~Q25	0.4671	0.7665	0.5873	0.4671
psycho=~Q8	NA	NA	NA	0.5500
psycho=~Q9	NA	NA	NA	0.2100
environment=~Q15	NA	NA	NA	0.2400

Source: [Population Models](#)

Table 2: Comparison of model variants used for sensitivity analysis

Metric	Meta-analytic (popmodel)	Naive (naivemodel)	Optimistic (optmodel)	H1 (h1model)
Loading Min	0.37	0.48	0.64	0.21
Loading Max	0.94	0.60	0.77	0.94
Mean Loading	0.75	0.51	0.69	0.70
Factor	0.20	0.30	0.08	0.20
Correlations				
Residual Min	0.12	0.64	0.41	0.12
Residual Max	0.86	0.77	0.59	0.67

Table 2: Comparison of model variants used for sensitivity analysis

Metric	Meta-analytic (popmodel)	Naive (naivemodel)	Optimistic (optmodel)	H1 (h1model)
Q3–Q4 Correlation	0.09	0.23	0.16	0.09

Source: [Population Models](#)

2.6 Analysis model

The analysis model (`analyzemodel`) serves dual purposes depending on the simulation paradigm:

- **Misspecification detection:** When data are generated from the misspecification model (`h1model`), the analysis model acts as the null hypothesis (H_0) by omitting the three cross-loadings. Power is then the probability of correctly rejecting this under-parameterized model.
- **Parameter recovery:** When data are generated from the meta-analytic, naive, or optimistic population models, the same structure is used to estimate the four-factor model with the *Q3–Q4* residual covariance, and power reflects the ability to recover individual parameters.

Key specification choices:

- **Identification:** Factor variances fixed to 1 (marker-free scaling), allowing all first-indicator loadings to be freely estimated.
- **Residual covariance:** The *Q3–Q4* covariance is freely estimated, which is both theoretically justified and empirically supported (Lin & Yao, 2022; Mosqueira Taípe et al., 2026).
- **Omitted cross-loadings:** The deliberate exclusion of cross-loadings for *Q8*, *Q9*, and *Q15* defines the misspecification that this tutorial’s main analysis is designed to detect.

```
# Factor loadings: freely estimated
psycho      =~ Q5 + Q6 + Q7 + Q11 + Q19 + Q26
physical    =~ Q3 + Q4 + Q10 + Q15 + Q16 + Q17 + Q18
social      =~ Q20 + Q21 + Q22
environment =~ Q8 + Q9 + Q12 + Q13 + Q14 + Q23 + Q24 + Q25

# Factor variances (=1) and free interfactor correlations
```

```

psycho      ~~ 1*psycho      + physical + social + environment
physical    ~~ 1*physical    + social   + environment
social      ~~ 1*social      + environment
environment  ~~ 1*environment

# Residual covariance Q3~~Q4: freely estimated
Q4 ~~ Q3

# Residual variances: freely estimated
Q5 ~~ Q5; Q6 ~~ Q6; Q7 ~~ Q7
Q11 ~~ Q11; Q19 ~~ Q19; Q26 ~~ Q26
Q3 ~~ Q3; Q4 ~~ Q4; Q10 ~~ Q10
Q15 ~~ Q15; Q16 ~~ Q16; Q17 ~~ Q17; Q18 ~~ Q18
Q20 ~~ Q20; Q21 ~~ Q21; Q22 ~~ Q22
Q8 ~~ Q8; Q9 ~~ Q9; Q12 ~~ Q12; Q13 ~~ Q13
Q14 ~~ Q14; Q23 ~~ Q23; Q24 ~~ Q24; Q25 ~~ Q25

```

Source: [Population Models](#)

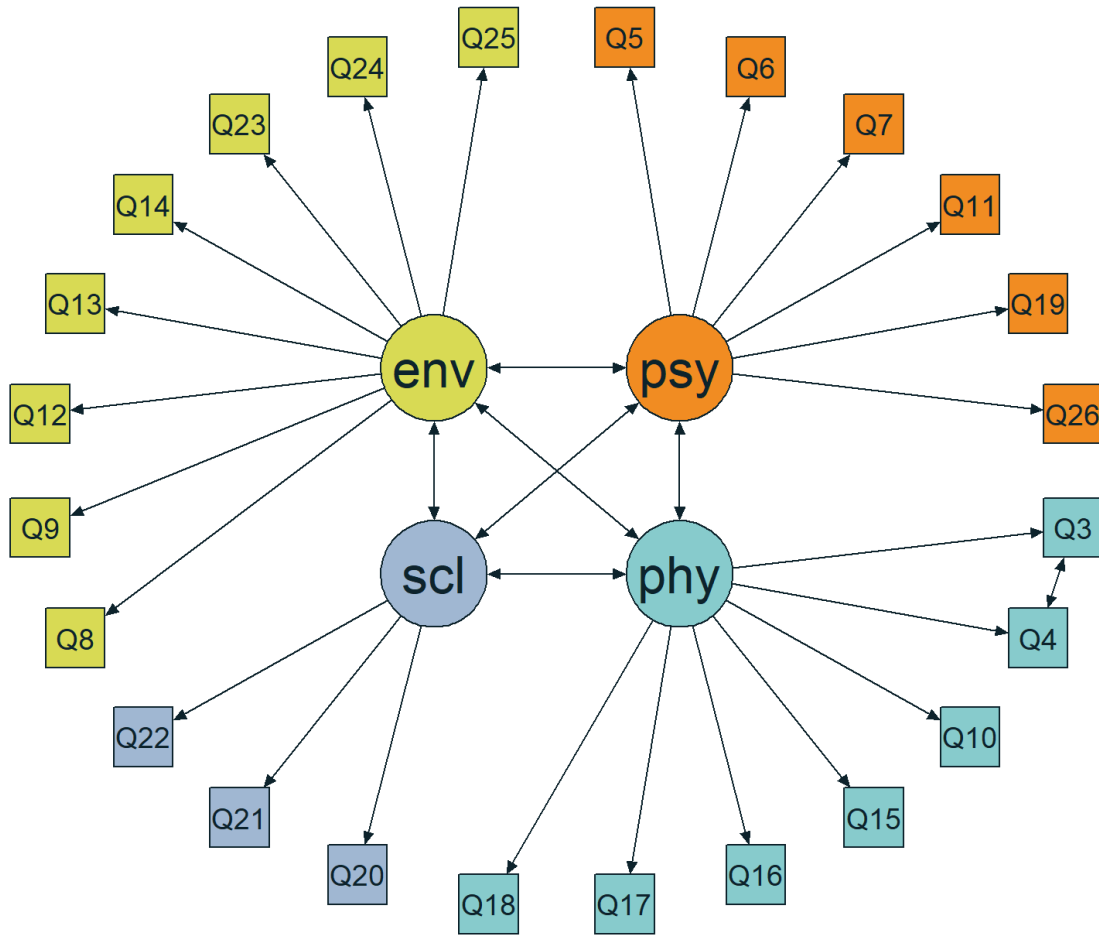


Figure 5: Path diagram of the analyzemodel

Source: [Population Models](#)

3 Analytical Power Analysis

Before running any data collection, analytical power analysis answers a fundamental planning question: how large must N be so that, if the model is wrong by at least a specified amount, we will reject it with acceptable probability? In SEM, the dominant analytical framework links misspecification to the non-central χ^2 distribution through the noncentrality parameter λ ,

which scales with both N and the magnitude of model discrepancy (Jak et al., 2021; MacCallum et al., 1996).

3.1 Model Degrees of Freedom

All analytical calculations in this section depend on the model degrees of freedom (df) — the difference between the number of unique variances and covariances in the observed data and the number of freely estimated parameters. This quantity enters the non-central χ^2 distribution directly and is therefore required before any a priori or post hoc power calculation can proceed:

Model degrees of freedom (via `semPower`): 249

Source: [Analytical Power Analysis](#)

3.2 A Priori: How Many Observations?

3.2.1 Model-free approach (RMSEA-based)

The simplest analytical strategy sets the target effect through the RMSEA, a noncentrality-based index that expresses misfit per degree of freedom and is therefore comparable across models of different complexity (MacCallum et al., 1996). Because RMSEA is directly linked to the population minimum of the fitting function through $F_0 = \text{RMSEA}^2 \times df$, specifying a target RMSEA is sufficient to define the entire noncentrality-based framework needed for the power calculation. The approach is “model-free” in the sense that no explicit population model needs to be specified: the required sample size depends only on RMSEA, α , desired power, and df (Jak et al., 2021; Jobst et al., 2023).

This strategy is most appropriate when the objective is to plan sample size for detecting overall model misspecification in a global chi-square test of fit, rather than for testing an individual parameter restriction (Moshagen & Bader, 2024; Wang, 2023).

`semPower: A priori power analysis`

F0	0.622500
RMSEA	0.050000
Mc	0.732531
GFI	0.950683
AGFI	0.940582

df	249
Required Num Observations	99
Critical Chi-Square	286.8077
NCP	61.00500
Alpha	0.050000
Beta	0.199217
Power (1 - Beta)	0.800783
Implied Alpha/Beta Ratio	0.250983

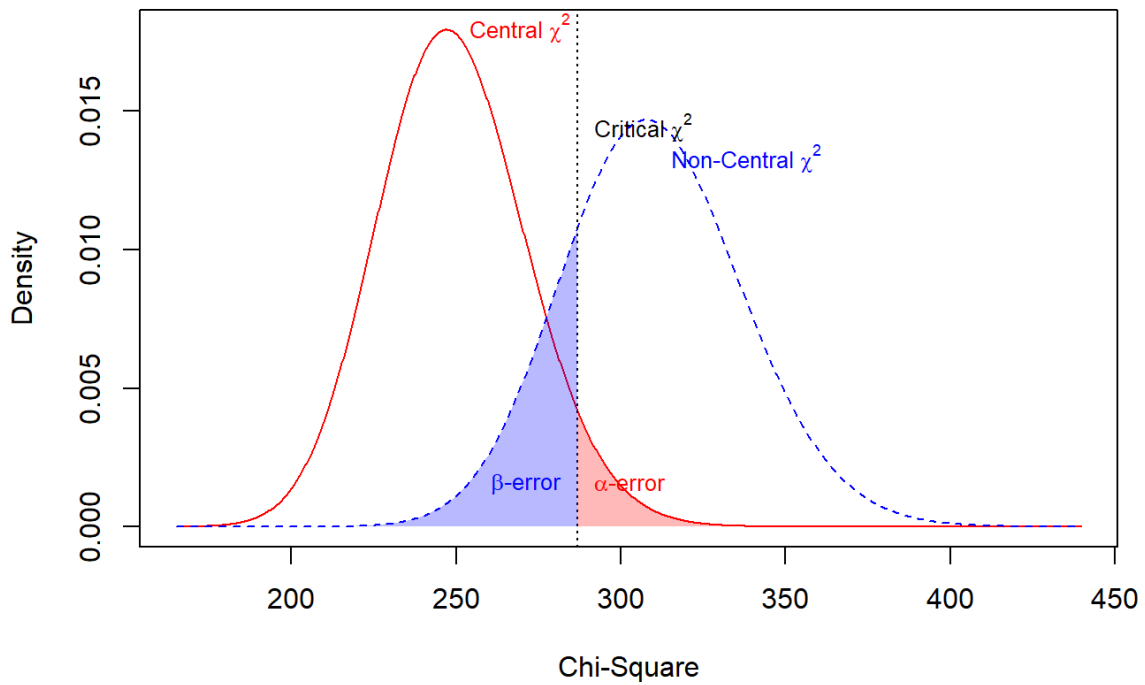


Figure 6: A priori power analysis for exact fit with RMSEA-based effect size

Source: [Analytical Power Analysis](#)

This output reports the minimum N required to detect misfit of $\text{RMSEA} = 0.05$ with power ≈ 0.80 at $\alpha = 0.05$. Note that 0.05 refers to the misspecification level to be detected under the alternative hypothesis, not to a tolerated misfit level under the null (Jak et al., 2021; MacCallum et al., 1996).

The output presents several equivalent representations of the same target effect. The value $F_0 = 0.623$ is the population minimum of the fitting function, obtained from $F_0 = df \cdot \text{RMSEA}^2 = 249 \cdot 0.05^2$. Alongside it, the summary also reports $Mc = 0.733$, $GFI = 0.951$, and $AGFI = 0.941$. These are not distinct effect sizes but alternative summaries of the same degree of model misfit, expressed in the metrics used by `semPower` (Moshagen & Bader, 2024).

The key conversion is:

$$\lambda = (N - 1)F_0 = (N - 1) df \text{RMSEA}^2$$

which translates the chosen misfit into the noncentrality parameter used to locate the alternative χ^2 distribution (Jak et al., 2021; MacCallum et al., 1996).

With $df = 249$, the a priori solution is $N = 99$. At that N , the noncentrality parameter is $\lambda = 61.005$ and the critical value is $\chi_c^2 = 286.808$. The implied $\beta \approx 0.199$ corresponds to power $1 - \beta \approx 0.801$, confirming that $N = 99$ is the minimum sample size at which the target operating characteristics are met (Jobst et al., 2023).

In Figure ??, the solid red curve is the central χ^2 under exact fit, the dashed blue curve is the noncentral χ^2 under the target misspecification, and the vertical dotted line at $\chi^2 = 286.808$ is the decision threshold (Jak et al., 2021; Moshagen & Bader, 2024).

3.2.2 Model-based approach

When substantive theory specifies the population structure, a model-based power analysis is more informative: rather than setting misfit to a generic RMSEA value, the effect is implied by the discrepancy between an explicitly defined population model and the restricted analysis model (Feng & Hancock, 2023; Wang, 2023). This makes the sample size estimate dependent on the hypothesized parameter configuration rather than on a conventional threshold.

`semPower.powerLav()` implements this by fitting the null model to the population-implied covariance matrix and translating the resulting discrepancy into F_0 and then into $\lambda = (N - 1)F_0$. When `modelH1` is omitted, the null model is evaluated against the saturated model, making the analysis a global test of fit rather than a test of any single restriction (Jobst et al., 2023; Moshagen & Bader, 2024). Because the effect is generated from the full model specification, any change to loadings, factor correlations, or residual variances will alter the required N — an important point when conducting sensitivity analyses (Buchberger et al., 2024).

`semPower: A priori power analysis`

<code>F0</code>	0.303870
<code>RMSEA</code>	0.034934

SRMR	0.175600
Mc	0.859044
GFI	0.975303
AGFI	0.970244
CFI	0.977582

df	249
Required Num Observations	202

Critical Chi-Square	286.8077
NCP	61.07791
Alpha	0.050000
Beta	0.198481
Power (1 - Beta)	0.801519
Implied Alpha/Beta Ratio	0.251914

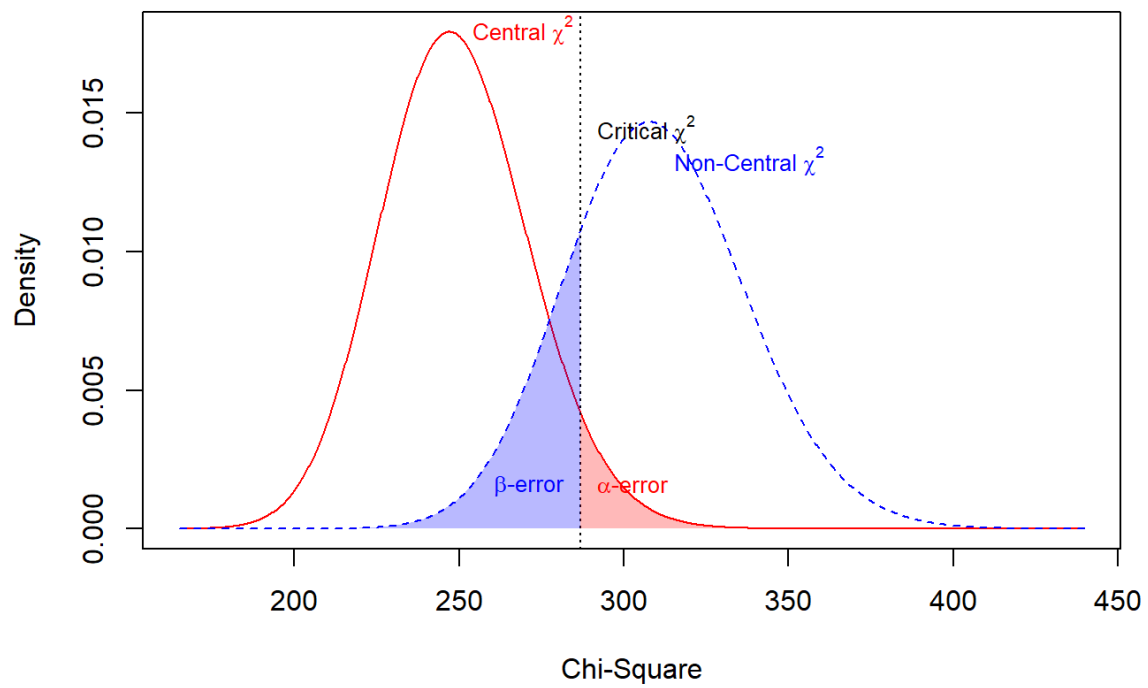


Figure 7: A priori power analysis Model-based approach (popModel)

Source: [Analytical Power Analysis](#)

The output reports a population discrepancy of $F_0 = 0.304$, equivalently expressed as RMSEA = 0.035 and $Mc = 0.859$. With $df = 249$, the required sample size is $N = 202$, and the corresponding critical chi-square is 286.808. The noncentrality parameter is $\lambda = 61.078$, consistent with $(N - 1)F_0$ at the obtained solution. The error rates are $\alpha = 0.05$, $\beta = 0.198$, and power = 0.802, confirming that the design conditions are met at $N = 202$.

3.3 Power Curves

Power curves visualize the relationship between sample size and the probability of detecting a prespecified degree of model misspecification, holding α and df constant. They serve as a planning aid by showing at a glance which N values are underpowered and where additional observations yield diminishing returns (Moshagen & Bader, 2024).

3.3.1 Model-free approach (RMSEA-based)

In the model-free setup, the power function depends entirely on RMSEA, df , and α — not on any substantive model parameters. Representing this function graphically answers a direct planning question: for a model with fixed df , how large must N be to achieve a sufficiently high probability of detecting misfit of at least the target magnitude (Jak et al., 2021; Jobst et al., 2023)? This visualization is especially useful when the research objective concerns global fit rather than individual parameter restrictions, and it also makes it easy to communicate design justification to reviewers or funding agencies (Moshagen & Bader, 2024; Wang, 2023).

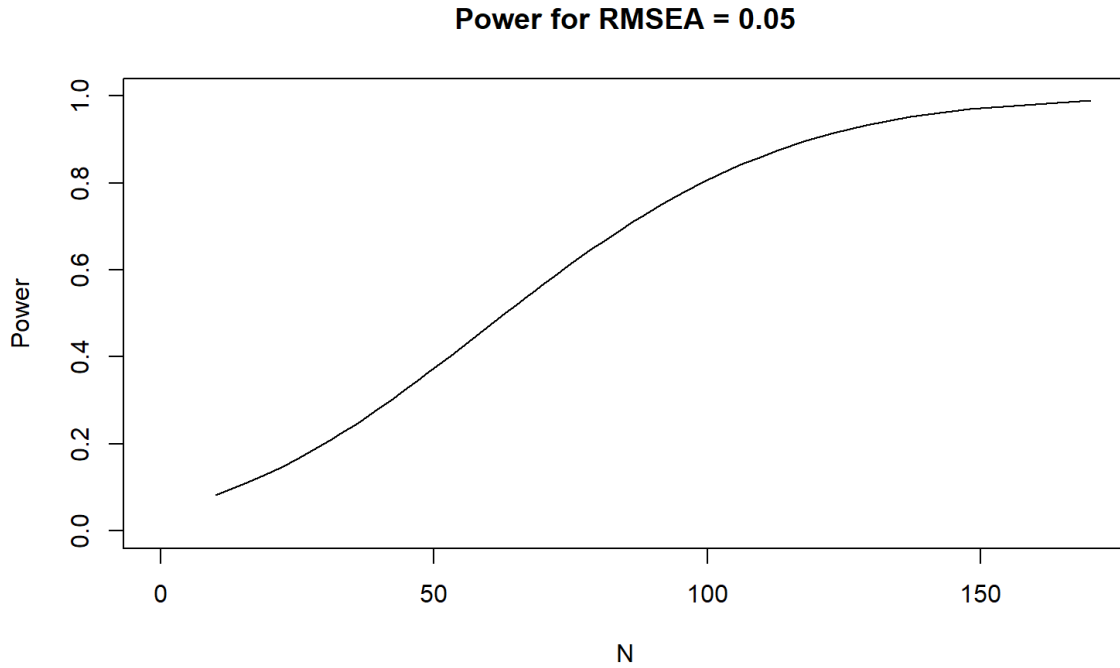


Figure 8: Power as a function of sample size for RMSEA = 0.05

Source: [Analytical Power Analysis](#)

Power increases monotonically with N . At the lower end of the range, the curve is well below the target threshold, indicating insufficient sensitivity to detect the prespecified misspecification; at higher N , it approaches the ceiling. The crossing point between the curve and the target power level identifies the minimum adequate sample size, and the shape of the curve shows how quickly additional observations convert into detection probability in this particular setting.

3.3.2 Model-based approach

When the effect is grounded in an explicitly specified population model, the power curve traces the same monotonic pattern but now reflects the particular covariance structure implied by the substantive hypotheses (Buchberger et al., 2024; Feng & Hancock, 2023). `semPower.powerPlot()` evaluates the previously specified model-based effect at each N in the grid, so the curve is a design-oriented summary of how sensitivity changes as sample size grows — without re-estimating any parameters (Moshagen & Bader, 2024).

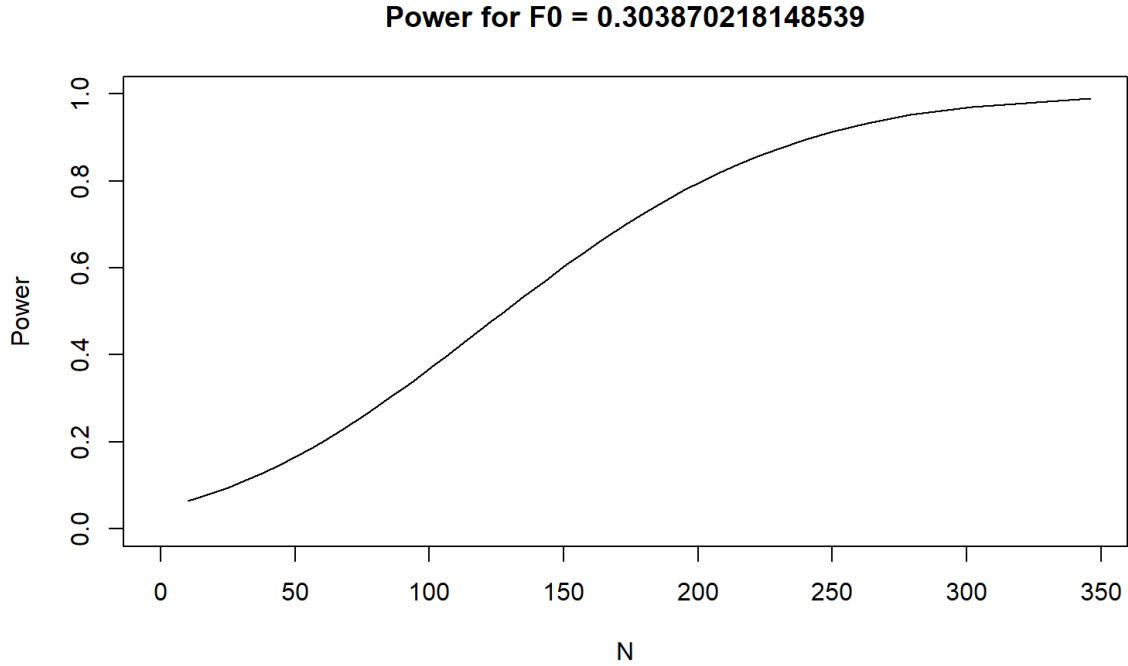


Figure 9: Power as a function of sample size for the population model (popModel)

Source: [Analytical Power Analysis](#)

The plot titled Power for $F_0 = 0.304$ shows a clear monotonic increase. At the lower end, power hovers near 0.07–0.10, indicating inadequate sensitivity for small samples. A period of rapid growth occurs between roughly $N = 120$ and $N = 220$, where each additional observation translates into a substantial gain. The conventional 0.80 threshold is crossed at approximately $N \approx 200$, and power approaches 0.95 around $N \approx 280$, with further gains becoming marginal above that.

3.4 Summary table

Table ?? shows that the model-based approach requires roughly twice as many observations as the model-free RMSEA-based approach to achieve equivalent target power. This gap arises because the model-based effect is derived from the actual discrepancy implied by the population covariance structure, rather than from a generic $\text{RMSEA} = 0.05$ convention; it therefore reflects a more substantively grounded — and in this case more conservative — estimate of the required N (Feng & Hancock, 2023; Wang, 2023).

Table 3: Summary of analytical power analysis results

Method	N Required	Power	Implied RMSEA	df
Model-free a priori (RMSEA = .05)	99	0.8008	0.0500	249
Model-based a priori (Population Model)	202	0.8015	0.0349	249

Source: [Analytical Power Analysis](#)

! Limitations of the Analytical Approach

The RMSEA/ χ^2 -based analytical method assumes continuous, multivariate normal data and Maximum Likelihood estimation. It can yield biased sample size estimates when distributions are non-normal, data are ordinal, or missing data reduce estimation efficiency. Furthermore, because power is framed around global misfit, the analytical approach provides no information about whether N is sufficient to detect specific weak loadings, factor correlations, or structural paths. These limitations motivate the simulation-based extensions developed in subsequent sections.

4 Monte Carlo Simulation: Ideal Conditions

Simulation-based power analysis in SEM operates on a different logic from its analytical counterpart. Rather than deriving power from the non-central χ^2 distribution, the approach repeatedly generates data from an assumed population model, fits the analysis model to each replicate, and summarizes rejection rates empirically. This makes the framework far more flexible: it can accommodate complex data conditions, evaluate multiple fit criteria simultaneously, and quantify estimation quality — bias, standard error accuracy, and coverage — alongside power (Muthén & Muthén, 2002; Pornprasertmanit et al., 2021).

The central distinction in the design is between the *data-generating model* and the *analysis model* (Feng & Hancock, 2023). The former defines the population from which replicated samples are drawn; the latter defines the structure that is repeatedly fit to each sample. Statistical power in SEM depends not only on N , but on the discrepancy between these two models, so this separation is essential (Feng & Hancock, 2023; Jobst et al., 2023).

In the main analysis, the reference condition uses data generated from `popmodel`, while the alternative condition uses data generated from `h1model` — a structurally equivalent model with three additional cross-loadings. The analysis model (`analyzemodel`) omits those cross-loadings, so power reflects the probability that global fit indices flag this deliberate misspecification. Varying N across a predefined sequence turns the simulation into a sample size planning

exercise: as N grows, the sampling distributions under the reference and alternative conditions separate, and the probability of correctly flagging the misspecified condition increases (Feng & Hancock, 2023; Jobst et al., 2023).

Power based on fit indices is therefore interpreted as the probability that a rejection rule — calibrated from the reference distribution at α — classifies data generated under the alternative as showing worse fit than expected under the well-fitting model. Practically, this shifts the inferential focus from testing a single parameter to evaluating whether global fit criteria are sensitive enough to detect a substantively meaningful form of misspecification across plausible sample sizes (Liang & Sterba, 2024; Wang, 2023).

4.1 Main analysis under ideal conditions

The Figure ?? and Figure ?? present the main results for the comparison between `popmodel` and `h1model` under ideal conditions (multivariate normality, no missing data).

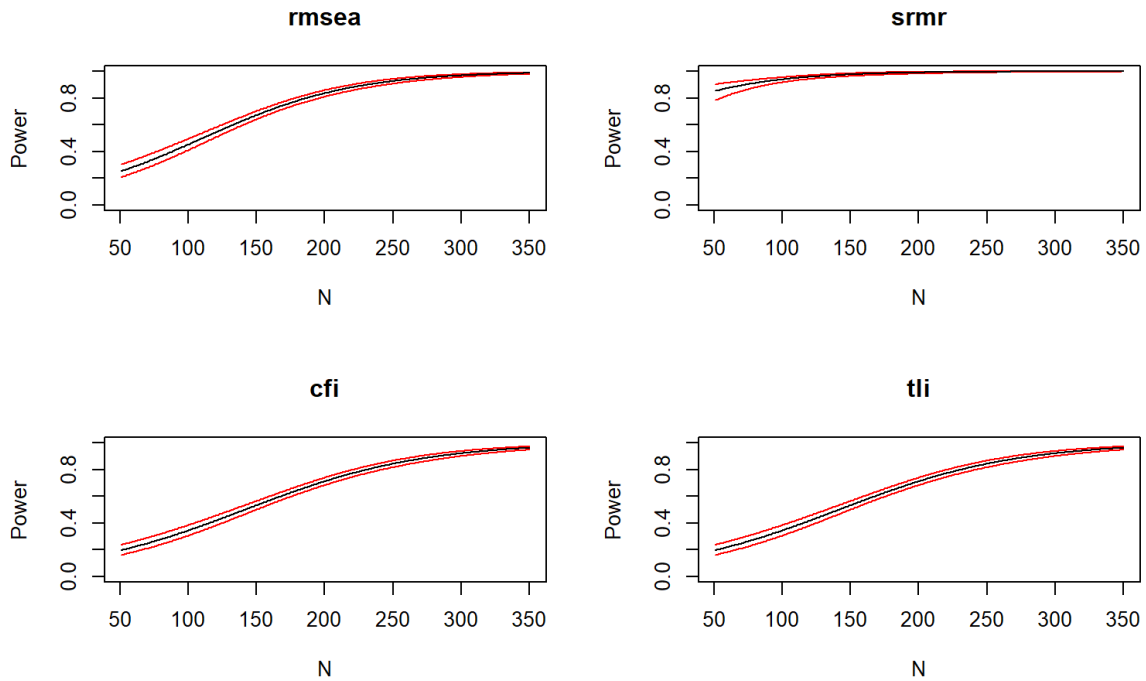


Figure 10: Power curves for fit indices under ideal conditions.

Source: [Simulation — Ideal Conditions](#)

Power increases monotonically with N for all four indices. At the lower end of the sample-size sequence, every index falls below the target threshold, indicating insufficient sensitivity. As N grows, the curves diverge: some indices reach the 0.80 target earlier than others, reflecting different degrees of sensitivity to the imposed cross-loading misspecification. The most conservative planning decision is therefore anchored in the last index to cross the target — the index whose detection requires the largest N .

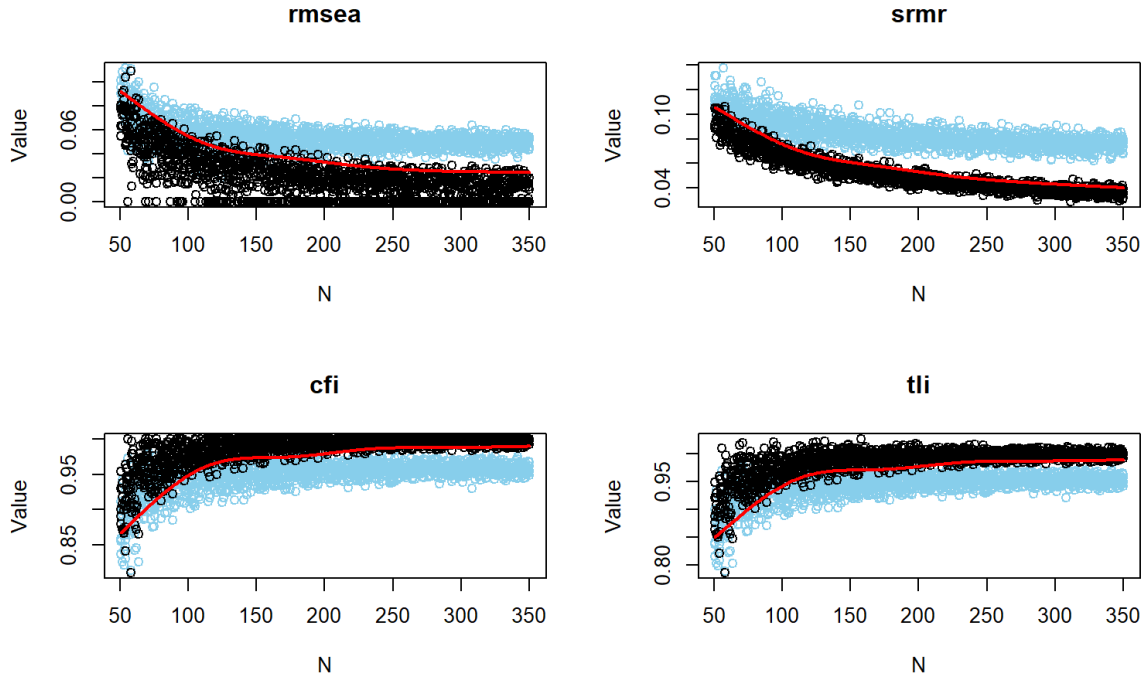


Figure 11: Power curves for fit indices under ideal conditions without logistic smoothing.

Source: [Simulation — Ideal Conditions](#)

The Figure ?? makes the discrimination mechanism visible. At small N , the sampling distributions of each fit index under the reference and alternative conditions overlap substantially, so the indices cannot reliably distinguish the two scenarios. As N increases, the alternative-condition distribution shifts toward worse fit while the reference distribution remains stable, widening the gap between them. This growing separation is the empirical basis for the power gains shown in the Figure ??, and it provides an intuitive picture of how fit indices work as decision variables rather than mere descriptive summaries (McNeish & Wolf, 2021; Pornprasertmanit et al., 2021).

Table 4: Power and fit index cutoff values under ideal conditions at $N = 231$.

	RMSEA	SRMR	CFI	TLI
Power	0.900	0.995	0.801	0.801
Cutoff	0.039	0.058	0.965	0.961

Source: [Simulation — Ideal Conditions](#)

At $N = 231$, power was 0.9 for RMSEA, 0.995 for SRMR, and 0.801 for both CFI and TLI. The ordering $\text{SRMR} > \text{RMSEA} > \text{CFI} \approx \text{TLI}$ is consistent across the power curves. The empirical cutoffs were 0.039 for RMSEA, 0.058 for SRMR, 0.965 for CFI, and 0.961 for TLI, defining the data-driven rejection boundaries at this N . Taken together, the results confirm that $N = 231$ is sufficient for all four indices to achieve at least 0.80 power in the ideal scenario, with SRMR and RMSEA providing a comfortable margin and CFI/TLI operating near the minimum threshold (Jobst et al., 2023; McNeish & Wolf, 2021).

4.2 Traditional cutoff values for fit indices

When traditional thresholds are applied — $\text{RMSEA} \leq .06$, $\text{CFI} \geq .95$, $\text{TLI} \geq .95$, $\text{SRMR} \leq .06$ — power is defined as the proportion of replications in which the misspecified model crosses those fixed boundaries in the direction of worse fit. This mirrors common applied practice, but a key limitation is that these thresholds were derived under specific simulation conditions and may not be universally valid across different model structures and sample sizes (Kyriazos, 2018; McNeish & Wolf, 2021).

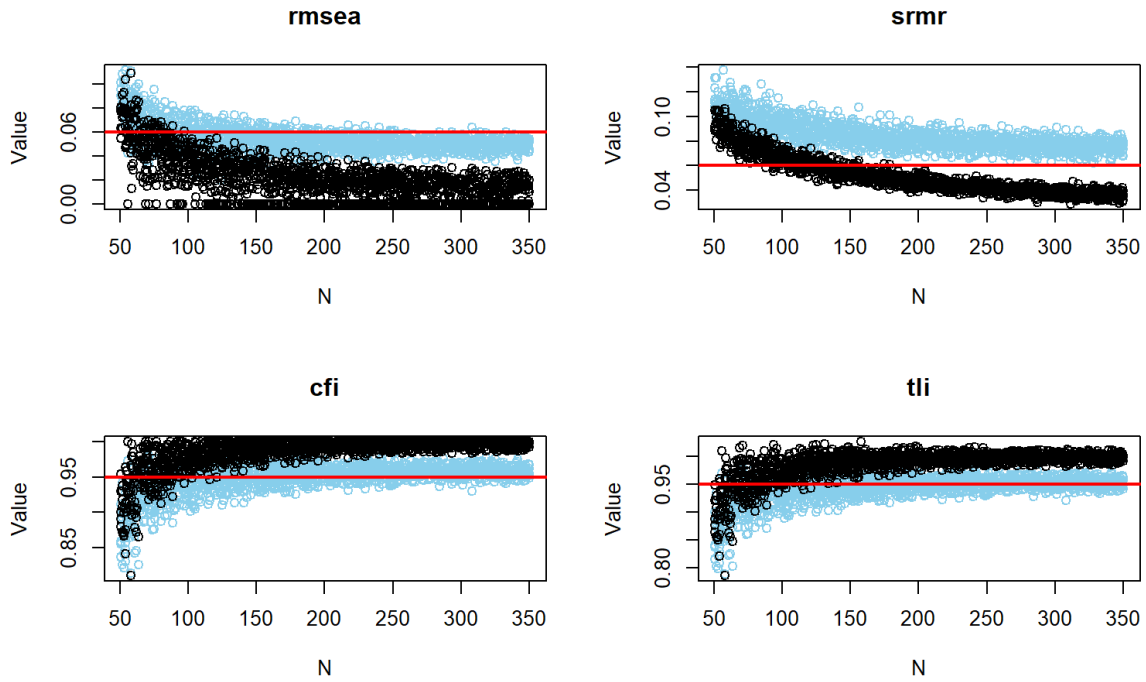


Figure 12: Power curves using traditional fit index cutoffs under ideal conditions.

Source: [Simulation — Ideal Conditions](#)

The contrast with the empirical-cutoff results is instructive. SRMR is the only index that achieves clear visual discrimination under the traditional threshold, and only at $N \approx 200$ or larger. RMSEA shows some separation but would require a cutoff near 0.04 rather than 0.06 to make a meaningful distinction in this figure. CFI and TLI show essentially no discrimination at the 0.95 threshold because both the reference and alternative point clouds cluster in the same region throughout the sample-size range (Figure ??). The Table ?? summarizes the power for each fit index under the traditional cutoff values.

Table 5: Power for traditional fit index cutoffs under ideal conditions at $N = 231$.

	CFI	TLI	RMSEA	SRMR
Power	0.381	0.579	0.153	> 0.98

Source: [Simulation — Ideal Conditions](#)

4.3 Targeting specific parameters

The previous analyses focused on global fit. However, power is always tied to a specific hypothesis, and different parameters within the same model can require markedly different N to reach the same target (Moshagen & Bader, 2024; Wang, 2023). This section evaluates power at the parameter level to identify which parts of the model are hardest to recover and should therefore anchor the sample size decision.

In `simsem`, parameter-level summaries include relative bias, standard-error bias, CI coverage, and detection power. These four indicators are not redundant: a parameter may show high power while still being poorly estimated, and estimation quality should always be verified alongside detection rates (Pornprasertmanit et al., 2021; Wang & Rhemtulla, 2021).

Table 6: Summary of parameter estimates under ideal conditions.

Table 6: Parameter estimates under ideal conditions {quarto-postprocess='true'}

Parameter	Rel Bias	Rel SE Bias	Coverage	Power (Not equal 0)
psycho=~Q5	-0.003	-0.043	0.947	1.000
psycho=~Q6	-0.005	-0.084	0.937	1.000
psycho=~Q7	-0.004	-0.066	0.940	1.000
psycho=~Q11	-0.005	-0.050	0.945	1.000
psycho=~Q19	-0.003	0.000	0.953	1.000
psycho=~Q26	-0.006	-0.079	0.938	1.000
physical=~Q3	-0.011	-0.063	0.947	1.000
physical=~Q4	-0.006	-0.066	0.945	1.000
physical=~Q10	-0.010	-0.062	0.947	1.000
physical=~Q15	0.068	-0.106	0.871	1.000
physical=~Q16	-0.011	-0.061	0.941	1.000
physical=~Q17	-0.010	-0.090	0.931	1.000
physical=~Q18	-0.011	-0.077	0.938	1.000
social=~Q20	-0.004	-0.050	0.941	1.000
social=~Q21	-0.003	-0.025	0.952	1.000
social=~Q22	0.002	-0.048	0.943	1.000
environment=~Q8	0.360	-0.090	0.520	0.997
environment=~Q9	0.105	-0.051	0.879	0.999
environment=~Q12	-0.012	-0.065	0.939	1.000
environment=~Q13	-0.010	-0.080	0.942	1.000
environment=~Q14	-0.009	-0.045	0.940	1.000
environment=~Q23	-0.010	-0.075	0.939	1.000
environment=~Q24	-0.006	-0.047	0.947	1.000
environment=~Q25	-0.011	-0.059	0.940	1.000
psycho~~physical	0.039	-0.069	0.941	0.751

Parameter	Rel Bias	Rel SE Bias	Coverage	Power (Not equal 0)
psycho~~social	-0.002	-0.069	0.935	0.727
psycho~~environment	0.286	-0.122	0.831	0.857
physical~~social	0.046	-0.047	0.944	0.760
physical~~environment	0.239	-0.091	0.875	0.839
social~~environment	0.038	-0.070	0.931	0.736
Q3~~Q4	0.007	-0.050	0.944	0.775
Q5~~Q5	-0.012	-0.043	0.944	0.999
Q6~~Q6	-0.017	-0.039	0.934	1.000
Q7~~Q7	-0.020	-0.071	0.929	0.993
Q11~~Q11	-0.014	-0.073	0.926	1.000
Q19~~Q19	-0.014	-0.050	0.933	1.000
Q26~~Q26	-0.018	-0.038	0.933	1.000
Q3~~Q3	-0.012	-0.059	0.935	1.000
Q4~~Q4	-0.015	-0.040	0.944	1.000
Q10~~Q10	-0.008	-0.044	0.934	1.000
Q15~~Q15	0.105	-0.068	0.896	1.000
Q16~~Q16	-0.011	-0.055	0.934	1.000
Q17~~Q17	-0.008	-0.084	0.934	0.999
Q18~~Q18	-0.008	-0.010	0.949	1.000
Q20~~Q20	-0.004	-0.091	0.943	0.848
Q21~~Q21	-0.015	-0.042	0.946	0.991
Q22~~Q22	-0.018	-0.050	0.933	1.000
Q8~~Q8	0.536	-0.058	0.136	1.000
Q9~~Q9	0.028	-0.059	0.944	1.000
Q12~~Q12	-0.001	-0.036	0.943	1.000
Q13~~Q13	0.004	-0.042	0.936	1.000
Q14~~Q14	-0.002	-0.085	0.937	1.000
Q23~~Q23	-0.001	-0.029	0.945	1.000
Q24~~Q24	-0.013	-0.055	0.937	1.000
Q25~~Q25	-0.007	-0.058	0.942	1.000

Source: [Simulation — Ideal Conditions](#)

Primary factor loadings for **psycho**, **physical**, and **social** recover very well: high power, minimal bias, modest SE distortion, and coverage close to nominal. Two exceptions stand out. The loading of *Q8* on **environment** and its residual variance are easily detected but not well recovered — the model would flag this parameter as non-zero while still estimating it with noticeable distortion. Item *Q15* shows a secondary warning on the same dimensions. Among the latent parameters, interfactor covariances — particularly those involving **social** and **environment** — are harder to recover than loadings, with lower power and weaker coverage. The residual covariance *Q3* ~~ *Q4* stands out as the most demanding residual parameter,

behaving more like a latent covariance than a measurement parameter in terms of inferential difficulty.

The required N for each parameter follows directly from these patterns (Jobst et al., 2023; Moshagen & Bader, 2024), as shown in Table ??.

Table 7: Sample size needed for 80% power under ideal conditions.

Parameter	N
environment $\sim$ Q8	51
environment $\sim$ Q9	51
psycho $\sim$ physical	200
psycho $\sim$ social	212
psycho $\sim$ environment	138
physical $\sim$ social	198
physical $\sim$ environment	148
social $\sim$ environment	208
Q3 $\sim$ Q4	181
Q5 $\sim$ Q5	51
Q7 $\sim$ Q7	51
Q17 $\sim$ Q17	51
Q20 $\sim$ Q20	143
Q21 $\sim$ Q21	51

Source: [Simulation — Ideal Conditions](#)

Many primary loadings and residual variances reach the target power at the low end of the simulated range. Latent covariances and the residual covariance $Q3 \sim Q4$ require substantially larger samples. In practice, the sample size decision should be driven by the hardest-to-detect parameters, not by the easiest ones — a principle that pushes the practical reference toward $N \approx 200$ rather than $N \approx 50$ (Pornprasertmanit et al., 2021; Wang & Rhemtulla, 2021). Specifically, the psycho $\sim$ social covariance requires $N = 212$, which sets the effective ceiling for parameter-level planning in this model.

4.4 Recovering population parameters

4.4.1 Parameter-level power under the naive model

Examining parameter recovery under the naive model serves as a worst-case benchmark: because this population uses the weakest loadings and the highest interfactor correlations, it maximizes the sample size needed to recover each parameter (Mosqueira Taipe et al., 2026). The goal here is not to evaluate global fit, but to identify the bottleneck parameter — the one that must

be detected if the entire set of model effects is to be reliably inferred (Pornprasertmanit et al., 2021; Wang & Rhemtulla, 2021).

Table 8: Parameter estimates and required sample size for 80% power under naive conditions.

Parameter	Rel Bias	Rel SE Bias	Coverage	Power (Not equal 0)	Required N
psycho~~physical	0.140	-0.944	0.913	0.793	180
psycho~~social	-0.009	-0.137	0.922	0.780	186
psycho~~environment	-0.003	-0.089	0.931	0.817	164
physical~~social	0.331	-0.971	0.924	0.775	192
physical~~environment	0.141	-0.932	0.929	0.793	178
social~~environment	-0.014	-0.123	0.928	0.780	184
Q3~~Q4	-0.013	-0.130	0.932	0.845	143

Source: [Simulation — Ideal Conditions](#)

Most factor loadings reach the 0.80 target with $N < 60$, confirming that the loading structure is not the binding constraint. The binding constraint is the set of latent covariances. Their required sample sizes are: psycho $\sim$ physical = 180, psycho $\sim$ social = 186, psycho $\sim$ environment = 164, physical $\sim$ social = 192, physical $\sim$ environment = 178, and social $\sim$ environment = 184. The residual covariance Q3 $\sim$ Q4 requires $N = 143$, placing it between the loadings and the latent covariances in difficulty. The largest required N is physical $\sim$ social = 192, which therefore governs the minimum sample size for the naive scenario.

4.4.2 Parameter-level power under the optimistic model

Under the optimistic model, stronger loadings and lower interfactor correlations substantially reduce sample size requirements, as expected (Mosqueira Taipei et al., 2026). The `findPower` results reveal a sparse set of finite required- N values: most parameters reach 0.80 power at very small samples, and only one parameter — the residual covariance Q3 $\sim$ Q4 — stands apart as the sole finite bottleneck (Lin & Yao, 2022).

Table 9: Parameter estimates and required sample size for 80% power under optimal conditions.

Parameter	Rel Bias	Rel SE Bias	Coverage	Power (Not equal 0)	Required N
Q3~~Q4	-0.008	-0.082	0.934	0.85	139

Source: [Simulation — Ideal Conditions](#)

The residual covariance Q3 $\sim$ Q4 requires $N = 139$, far above the requirements for all other listed parameters. This value alone governs the sample size decision under the optimistic

scenario, since every other finite parameter is adequately powered at $N = 51$. The contrast between the naive and optimistic results demonstrates how sensitive N recommendations are to the assumed population: the same model structure implies very different requirements depending on which parameter set is used, reinforcing the importance of sensitivity analyses across population variants (Buchberger et al., 2024; Irmer et al., 2024).

4.5 Sensitivity to population model assumptions

4.5.1 Fit-index-based power under the naive population

Changing the reference population model from `popmodel` to `naivemodel` tests how sensitive the main conclusions are to the assumed parameter values. Under this scenario, the cutoff values are derived from the naive-model null distribution, and power is the proportion of `h1model`-generated replications that exceed those cutoffs. Because the naive model implies weaker signal, larger samples are required before the fit indices can reliably distinguish the reference from the misspecified condition (Buchberger et al., 2024).

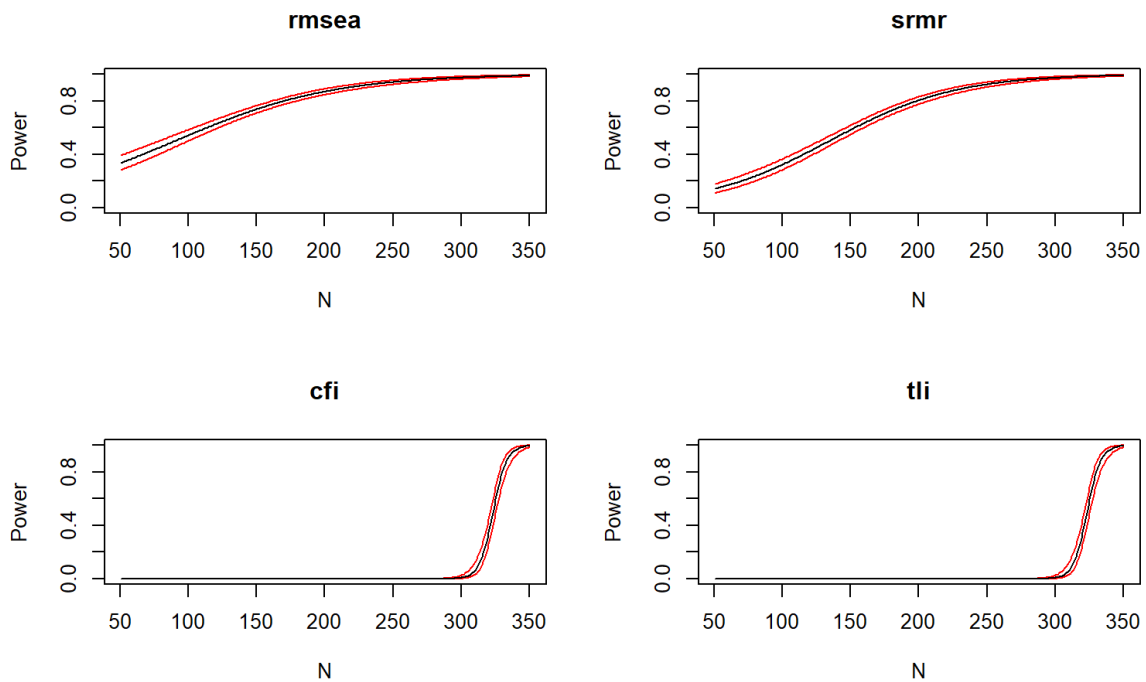


Figure 13: Power curves for fit indices using the naive population model.

Source: [Simulation — Ideal Conditions](#)

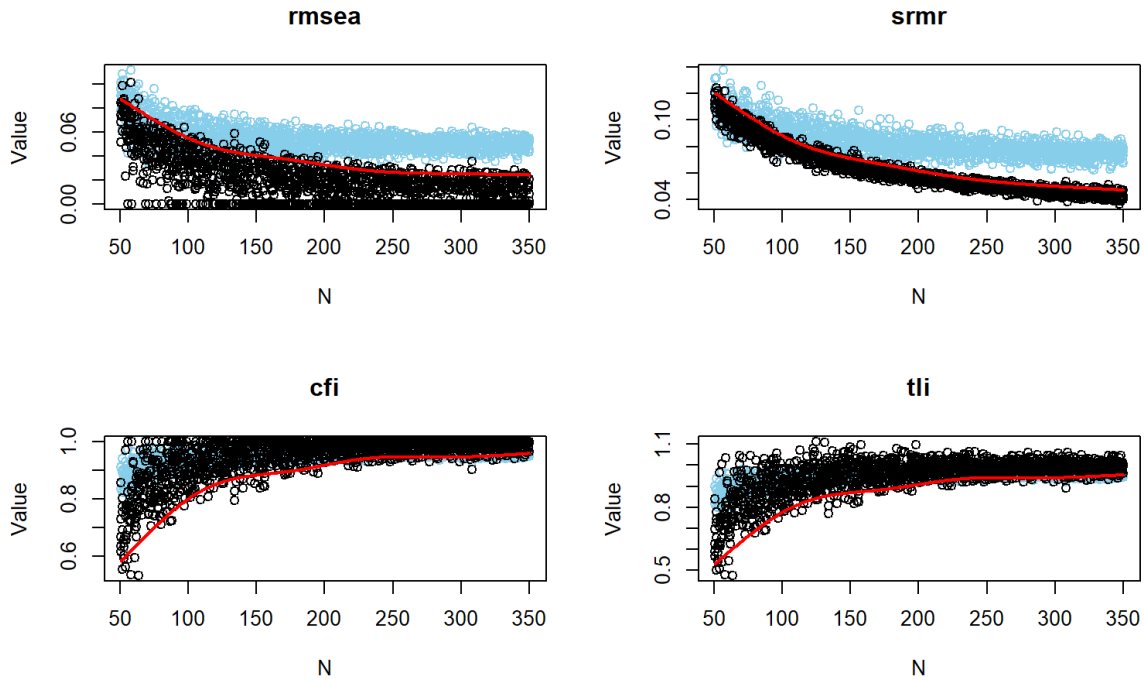


Figure 14: Power curves for fit indices using the naive population model without logistic smoothing.

Source: [Simulation — Ideal Conditions](#)

The power curves show two distinct patterns. RMSEA and SRMR increase gradually from moderate levels at $N = 50$ and approach 1.00 as sample size grows. CFI and TLI remain close to zero across most of the range, then rise sharply only near $N \approx 310$ – 340 . In the raw fit-index plots, the two point clouds are clearly separated for RMSEA and SRMR at $N > 200$, but overlap substantially for CFI and TLI at all but the largest sample sizes.

Table 10: Power and fit index cutoff values using the naive population model at $N = 330$.

	RMSEA	SRMR	CFI	TLI
Power	0.984	0.986	0.801	0.801
Cutoff	0.022	0.047	0.964	0.959

Source: [Simulation — Ideal Conditions](#)

At $N = 330$, all four indices exceed 0.80, but the margin differs substantially. RMSEA and SRMR are near 0.99; CFI and TLI just clear the threshold. The empirical cutoffs at this N are

more demanding than conventional thresholds, reflecting the specific distribution of the null model (McNeish & Wolf, 2021). The pattern is consistent with the main analysis: residual-type and absolute indices detect the misspecification more readily than incremental indices, though here the gap is considerably wider because the naive population reduces the overall signal available to CFI and TLI.

4.5.2 Sensitivity under the optimistic population

Under the optimistic population, the same logic applies but the baseline signal is stronger, so power accumulates more rapidly and at smaller N values than in the naive scenario (Mosqueira Taípe et al., 2026). Cutoffs are derived from the optimistic null distribution, and power is evaluated against `h1model`-generated data using the same fit indices.

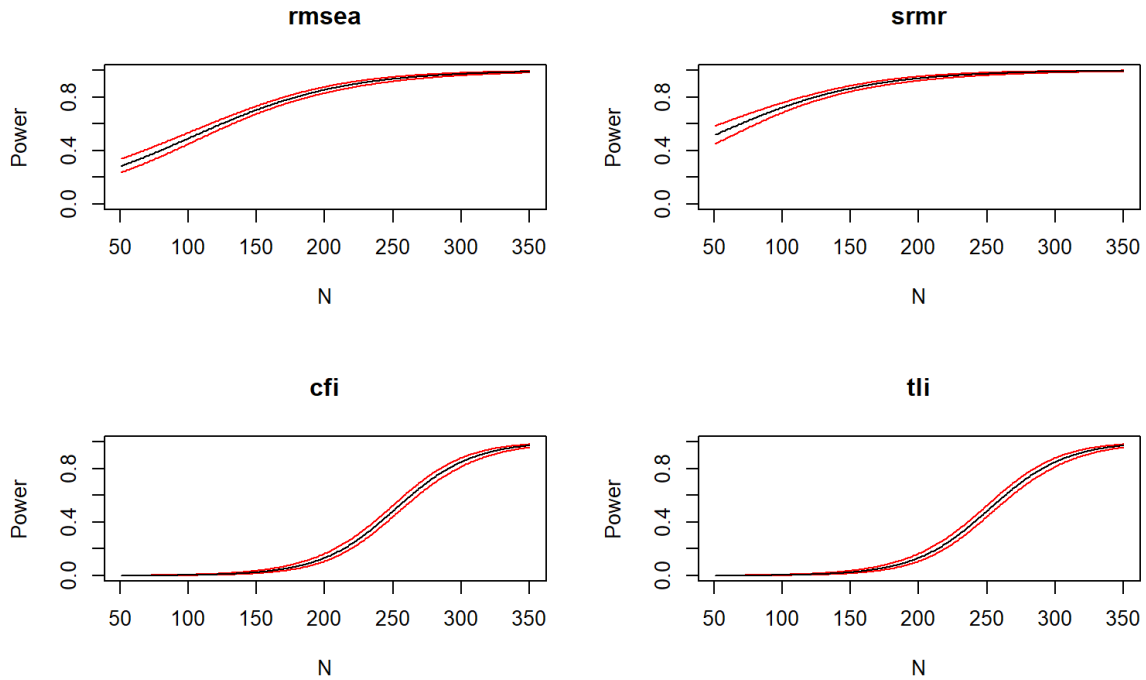


Figure 15: Power curves for fit indices using the optimal population model.

Source: [Simulation — Ideal Conditions](#)

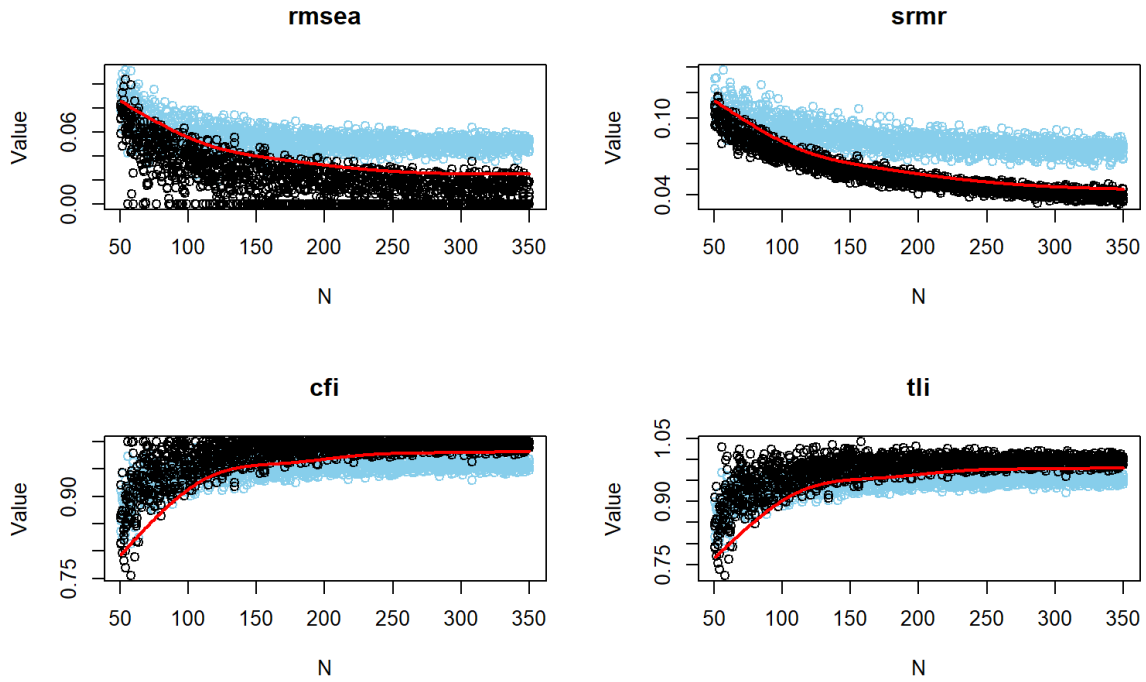


Figure 16: Power curves for fit indices using the optimal population model without logistic smoothing.

Source: [Simulation — Ideal Conditions](#)

The power curves mirror the qualitative pattern seen under the naive population — SRMR is the strongest index, followed by RMSEA, with CFI and TLI rising later and more steeply — but all curves shift leftward, reaching target power at smaller N values. The unsmoothed plots confirm that RMSEA and SRMR show clearer separation earlier than CFI and TLI, with visible overlap at lower and intermediate sample sizes for the incremental indices.

Table 11: Power and fit index cutoff values using the optimal population model at $N = 291$.

	RMSEA	SRMR	CFI	TLI
Power	0.968	0.988	0.805	0.805
Cutoff	0.028	0.051	0.970	0.966

Source: [Simulation — Ideal Conditions](#)

At $N = 291$, all four indices exceed 0.80. SRMR and RMSEA are well above the benchmark; CFI and TLI are adequate but comparatively limited. The empirical cutoffs at $N = 291$ are

again more conservative than conventional thresholds, and the gap between index sensitivities is smaller here than under the naive population. Taken together, the naive ($N = 330$) and optimistic ($N = 291$) results bracket the main estimate ($N = 231$), providing a concrete picture of how population uncertainty translates into uncertainty about required sample size.

5 Monte Carlo Simulation: Realistic Scenarios

The ideal-condition analyses establish a theoretical baseline, but actual data collection rarely conforms to multivariate normality with complete observations. When non-normality or missing data are anticipated, analytical power estimates may be optimistic, and the required N should be reassessed under more realistic assumptions (Feng & Hancock, 2023; Wang, 2023). Monte Carlo simulation is especially valuable here because it can encode precisely the data conditions expected in practice — including the missing-data mechanism, the distributional shape, and the estimator — and propagate them through to the power estimate (Irmer et al., 2024; Schoemann et al., 2014).

The goal of this section is not to replace the ideal-condition results but to probe how sensitive they are to two common violations: missing data and non-normality. Both analyses follow the same design logic as before — reference data from `popmodel`, alternative data from `h1model`, analysis model `analyzemodel` — so the results remain directly comparable (Jobst et al., 2023; Moshagen & Bader, 2024).

5.1 Missing Data (MCAR)

Missing data reduce the amount of usable information per participant and therefore generally lower estimation efficiency and statistical power, even when data are missing completely at random (MCAR) and the analysis model handles missingness correctly via full information maximum likelihood (Jak et al., 2021; Wolf et al., 2013). The relevant design question is not only how many cases are planned but how much information survives the anticipated missing-data process (Feng & Hancock, 2023).

In this simulation, 10% of observations are set missing at random prior to each analysis. All other aspects of the design are unchanged from the ideal-condition analysis.

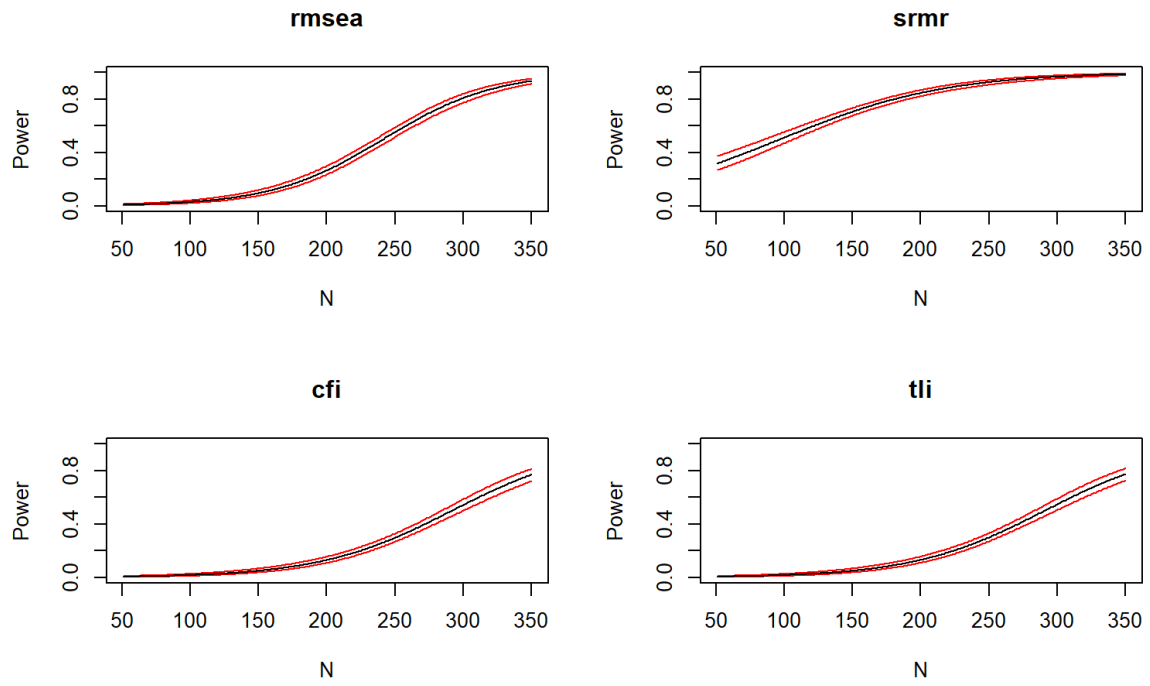


Figure 17: Power curves for fit indices under missing data conditions.

Source: [Simulation — Realistic Scenarios](#)

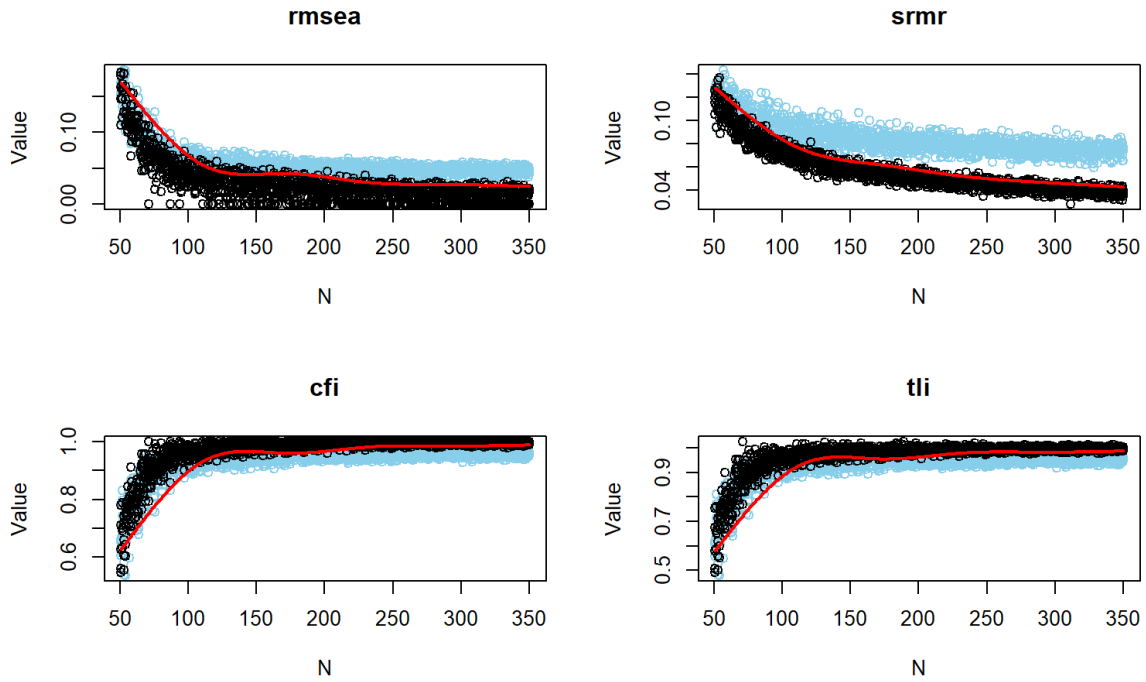


Figure 18: Power curves for fit indices under missing data conditions without logistic smoothing.

Source: [Simulation — Realistic Scenarios](#)

Both figures show the expected pattern: power curves shift rightward relative to the ideal condition, requiring larger N to reach the same thresholds. The separation between distributions in the raw fit-index plots is less pronounced, especially for CFI and TLI, which show more overlap at small and intermediate sample sizes.

Table 12: Power and fit index cutoff values under missing data conditions at $N = 359$.

	RMSEA	SRMR	CFI	TLI
Power	0.947	0.987	0.802	0.806
Cutoff	0.011	0.035	1.008	1.010

Source: [Simulation — Realistic Scenarios](#)

At $N = 359$, power was 0.947 for RMSEA, 0.987 for SRMR, 0.802 for CFI, and 0.806 for TLI. SRMR remains the strongest index and RMSEA is close behind; CFI and TLI are adequate but marginally so. The empirical cutoffs were 0.011 for RMSEA, 0.035 for SRMR, 1.008 for CFI, and 1.01 for TLI. The values above 1.00 for CFI and TLI indicate that the classification

boundary from the simulation falls at the upper edge of the index scale, which explains why their power is only marginally above 0.80 at this N (McNeish & Wolf, 2021). Comparing $N = 231$ (ideal) with $N = 359$ (10% MCAR) makes the information cost of missing data concrete: approximately 128 additional participants are needed to restore the same level of detection performance.

5.2 Non-Normality

Mild departures from normality — skewness in the range $[-1, +1]$ and excess kurtosis in the range $[1, 2]$ — are common in applied social science measurement and can affect the behavior of fit indices even when robust estimators are used (Irmer et al., 2024; Kyriazos, 2018). This analysis evaluates how such departures alter the power conclusions relative to the ideal condition.

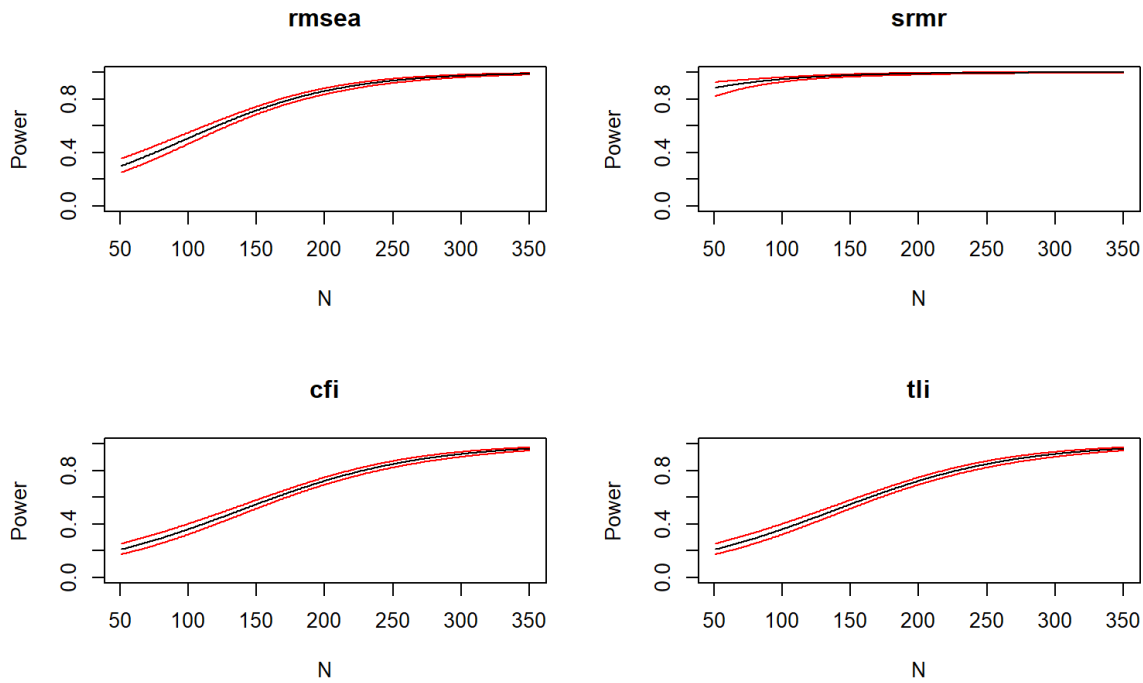


Figure 19: Power curves for fit indices under non-normal conditions.

Source: [Simulation — Realistic Scenarios](#)

The ordering of sensitivity is stable: $\text{SRMR} > \text{RMSEA} > \text{CFI} \approx \text{TLI}$. SRMR reaches near-ceiling power early in the sample-size range; RMSEA follows steadily; CFI and TLI rise more slowly and remain below the other two across most of the evaluated range.

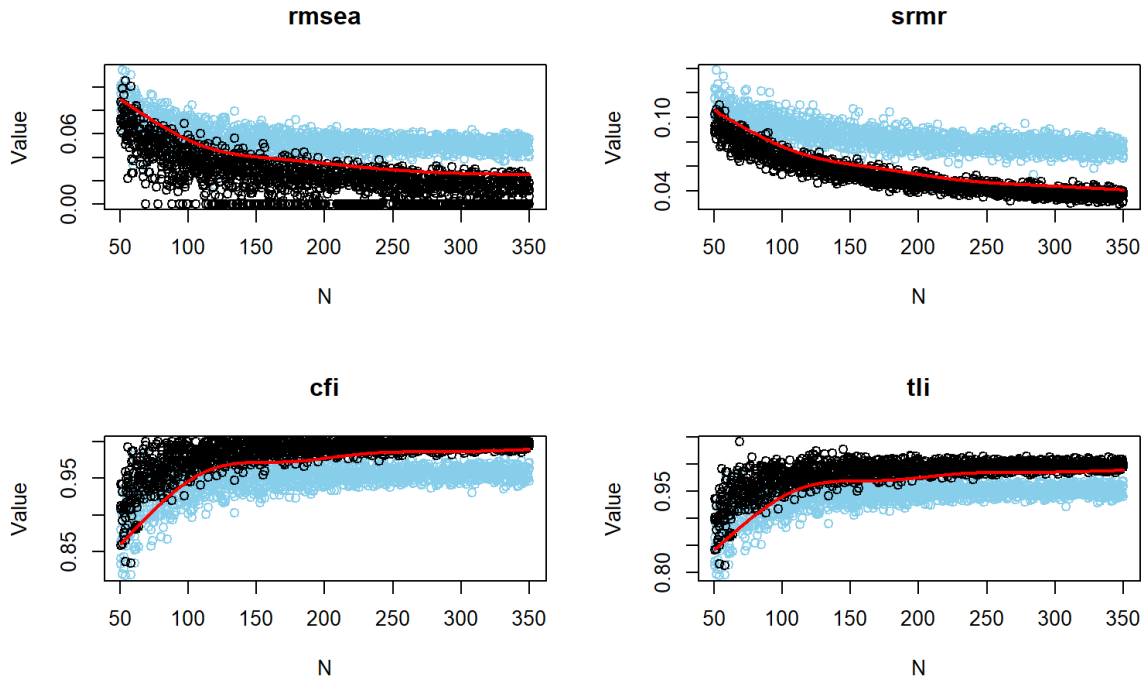


Figure 20: Power curves for fit indices under non-normal conditions without logistic smoothing.

Source: [Simulation — Realistic Scenarios](#)

The raw fit-index plots confirm the ordering. For RMSEA and SRMR, the two point clouds are increasingly distinct as N grows; for CFI and TLI, overlap is more visible at lower and intermediate sample sizes and clears only at larger N .

Table 13: Power and fit index cutoff values under non-normal conditions at $N = 228$.

	RMSEA	SRMR	CFI	TLI
Power	0.91	0.994	0.800	0.80
Cutoff	0.04	0.059	0.964	0.96

Source: [Simulation — Realistic Scenarios](#)

At $N = 228$, power was 0.91 for RMSEA, 0.994 for SRMR, 0.8 for CFI, and 0.8 for TLI. Notably, this result is almost identical to the ideal-condition estimate of $N = 231$, suggesting that mild non-normality introduces negligible additional cost under the current design and model structure (Irmer et al., 2024). The empirical cutoffs at $N = 228$ were 0.04 (RMSEA),

0.059 (SRMR), 0.964 (CFI), and 0.96 (TLI), consistent with the discrimination patterns in the figures.

5.3 Simulation with fixed N

Having established minimum N estimates under various conditions, the question becomes whether a single, fixed sample size performs reliably as a design choice. Fixing $N = 350$ — a conservative value anchored above the most demanding simulation result ($N = 330$ under the naive population) — allows us to ask: does this N consistently separate the reference and misspecified models across many independent replications, including under realistic data conditions (10% MCAR + mild non-normality)?

This step also highlights the practical difference between empirical and traditional fit criteria. Empirical cutoffs are calibrated to the actual sampling distribution of the fitted model; traditional cutoffs are generic conventions. The two approaches can lead to very different conclusions about adequacy (McNeish & Wolf, 2021; Pornprasertmanit et al., 2021).

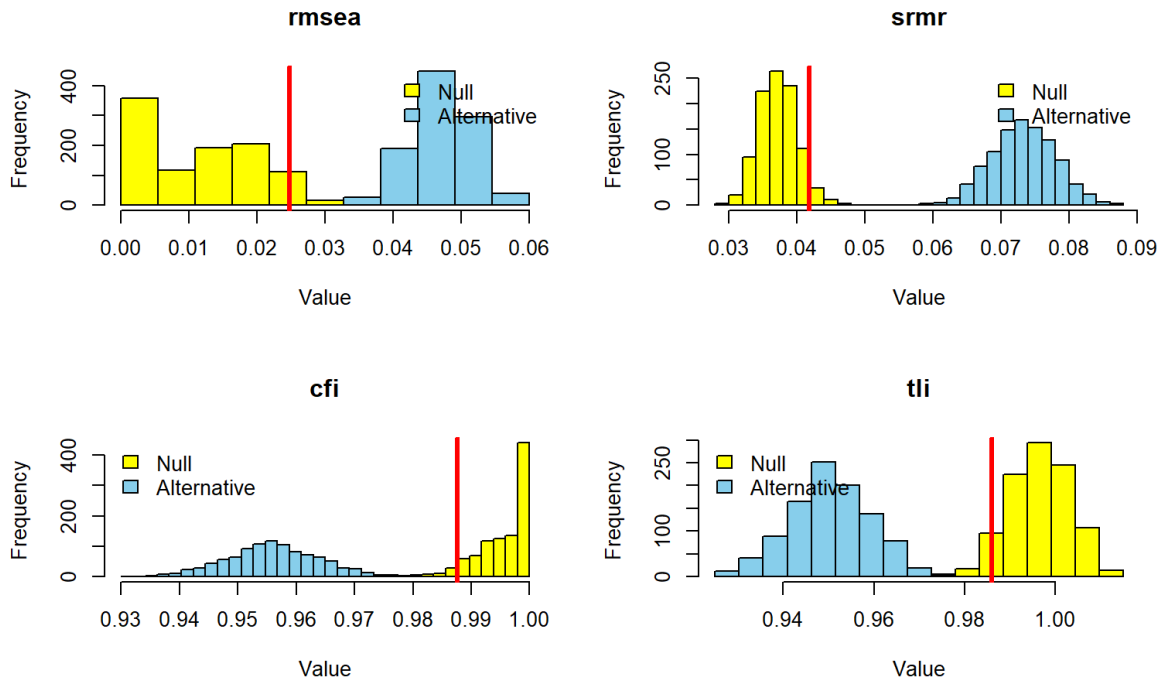


Figure 21: Power curves for fit indices with fixed sample size.

Source: [Simulation — Fixed N](#)

With $N = 350$, the main analysis achieves power = 1.000 for all four fit indices using empirical cutoffs. The distributions of the reference and alternative conditions are completely separated in every panel, confirming that this sample size provides essentially perfect discrimination under the target design (Liang & Sterba, 2024; Moshagen & Bader, 2024). The empirical cutoffs at $N = 350$ serve as data-driven benchmarks for this specific population-analysis model combination, and their values (Table ??) differ noticeably from conventional thresholds — which underscores the value of deriving rejection rules from the simulation rather than from generic recommendations.

Table 14: Power and fit index cutoff values with fixed sample size.

	TRADITIONALCHI	RMSEA	SRMR	CFI	TLI
Power	1.000	1.000	1.000	1.000	1.000
Cutoff	0.025	0.042	0.988	0.986	0.025

Source: [Simulation — Fixed N](#)

5.3.1 Power analysis with traditional cutoffs

Applying traditional rule-of-thumb thresholds to the same fixed- N simulation produces a strikingly different picture. Power remains high for SRMR, falls substantially for CFI and TLI, and drops to zero for RMSEA — not because the sample is inadequate, but because the conventional $\text{RMSEA} \leq 0.05$ threshold is misaligned with the empirical distribution of this specific model (Kyriazos, 2018; McNeish & Wolf, 2021). This contrast is one of the clearest illustrations in this tutorial of why power conclusions depend on how the rejection criterion is defined, not only on N .

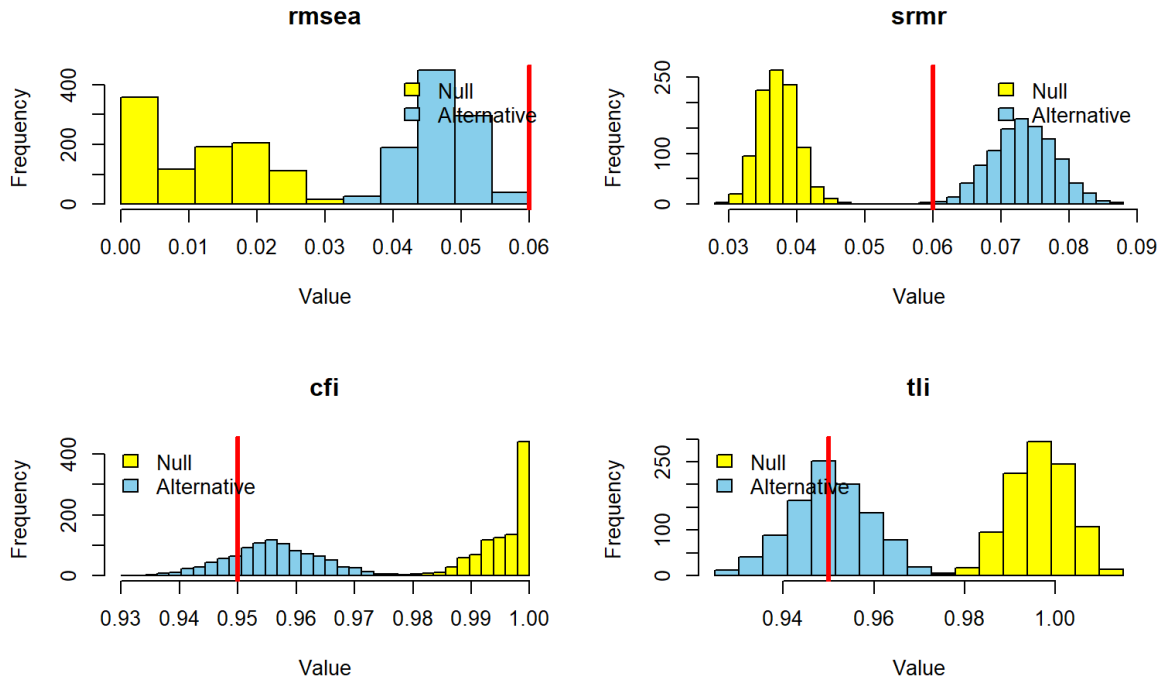


Figure 22: Power curves using traditional fit index cutoffs with fixed sample size.

Source: [Simulation — Fixed N](#)

Table 15: Power for traditional fit index cutoffs with fixed sample size.

	CFI	TLI	RMSEA	SRMR
Power	0.222	0.483	0	0.997

Source: [Simulation — Fixed N](#)

Taken together, the fixed- N results support $N = 350$ as a conservative, stable design choice for the main comparison: it delivers perfect discrimination when empirical cutoffs are used, and its failures under traditional cutoffs reveal a limitation of those cutoffs rather than of the design itself (Liang & Sterba, 2024; Pornprasertmanit et al., 2021).

6 Post Hoc Power Analysis

Post hoc power analysis evaluates the detection capacity of a study design after the data have been collected, using the observed N , the fitted model, and a specified reference structure. In

SEM, this is meaningful only with respect to a clearly defined comparison: power is always the probability of rejecting a specific model under a specific alternative, not a general property of a dataset or a software routine (Jobst et al., 2023; Moshagen & Bader, 2024). In this section, we evaluate post hoc power against an empirical dataset from Mendeley (Rogers, 2022), fitting `analyzemodel` to real WHOQOL-BREF data and assessing how reliably the collected N would have rejected meaningful misspecification.

6.1 With Simulation

The simulation-based strategy first fits the analysis model to the real data to obtain an empirically grounded reference structure. It then generates data from both the null model and the alternative model at the observed N and quantifies how well the fit indices separate the two conditions. This approach is particularly informative when the effect of interest is defined in model terms, because the full parameterization — loadings, residual structure, latent correlations — shapes the discrepancy and therefore the power estimate (Feng & Hancock, 2023; Wang & Rhemtulla, 2021).

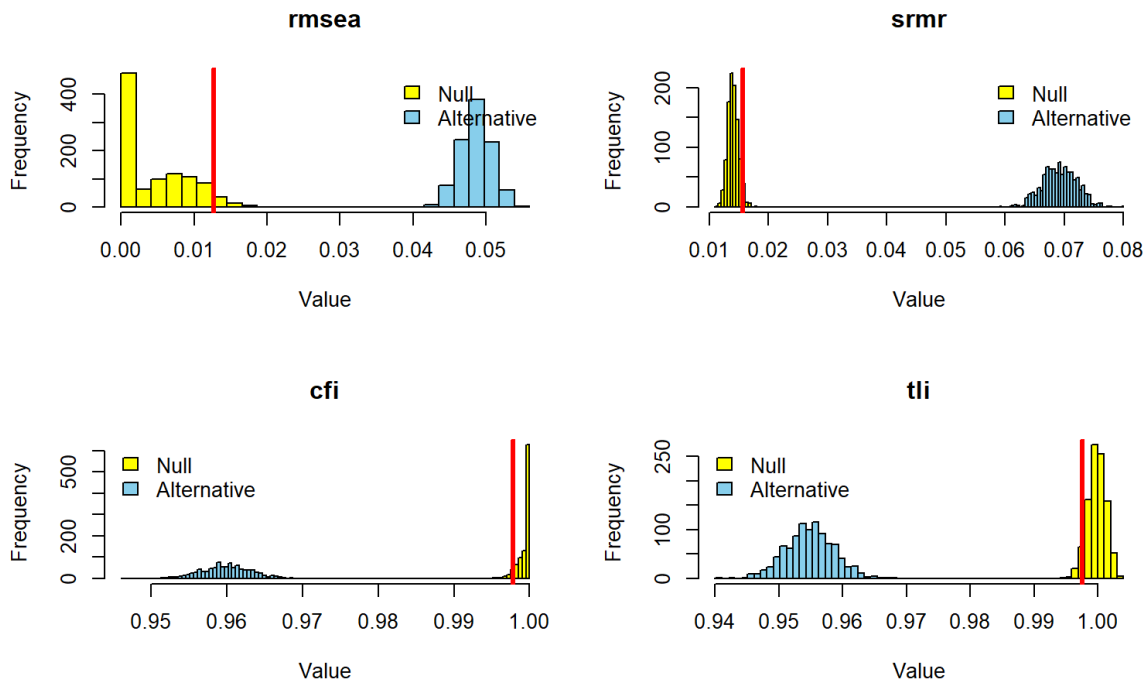


Figure 23: Power curve for post hoc analysis based on simulation.

Source: [Post Hoc Power Analysis](#)

All four fit indices reached power equal to 1.00. In the figure, every panel shows complete separation between the null and alternative distributions, with the red cutoff line falling cleanly between them and no visible overlap. For RMSEA, the null distribution concentrates near zero while the alternative is centered around 0.05. For SRMR, the null clusters near 0.01–0.02 and the alternative near 0.06–0.08. For CFI and TLI, the null is close to 1.00 and the alternative centers around 0.95–0.97. The complete discrimination in every panel confirms that the observed N would have been fully adequate to detect the target misspecification (Pornprasertmanit et al., 2021). Note that some indices are operating below their conventional thresholds, which is a consequence of the empirical cutoffs derived from the null distribution in this specific setting — not an indication that model fit is poor.

6.2 With Analytical Solutions

Analytical post hoc power expresses the effect through F_0 or RMSEA and links it to the noncentrality-based framework of the likelihood-ratio test. This is computationally immediate and free from Monte Carlo sampling error, making it the preferred first-pass approach when the asymptotic assumptions underlying the chi-square test are credible (Jak et al., 2021; Moshagen & Bader, 2024). Three specifications are compared here, illustrating a spectrum from the most generic to the most substantively specific formulation.

6.2.1 Model-free specification

The first specification supplies $\text{RMSEA} = 0.05$ directly, treating misspecification as a generic level of global misfit per degree of freedom. This makes the analysis comparable across SEM families sharing the same df , but it abstracts away from the particular covariance structure of the WHOQOL-BREF model (Jak et al., 2021; MacCallum et al., 1996).

`semPower: Post hoc power analysis`

<code>F0</code>	0.622500
<code>RMSEA</code>	0.050000
<code>Mc</code>	0.732531
<code>GFI</code>	0.950683
<code>AGFI</code>	0.940582
<code>df</code>	249
<code>Num Observations</code>	1047
<code>NCP</code>	651.1350
<code>Critical Chi-Square</code>	286.8077

Alpha	0.050000
Beta	4.771426e-48
Power (1 - Beta)	> 0.9999
Implied Alpha/Beta Ratio	1.047905e+46

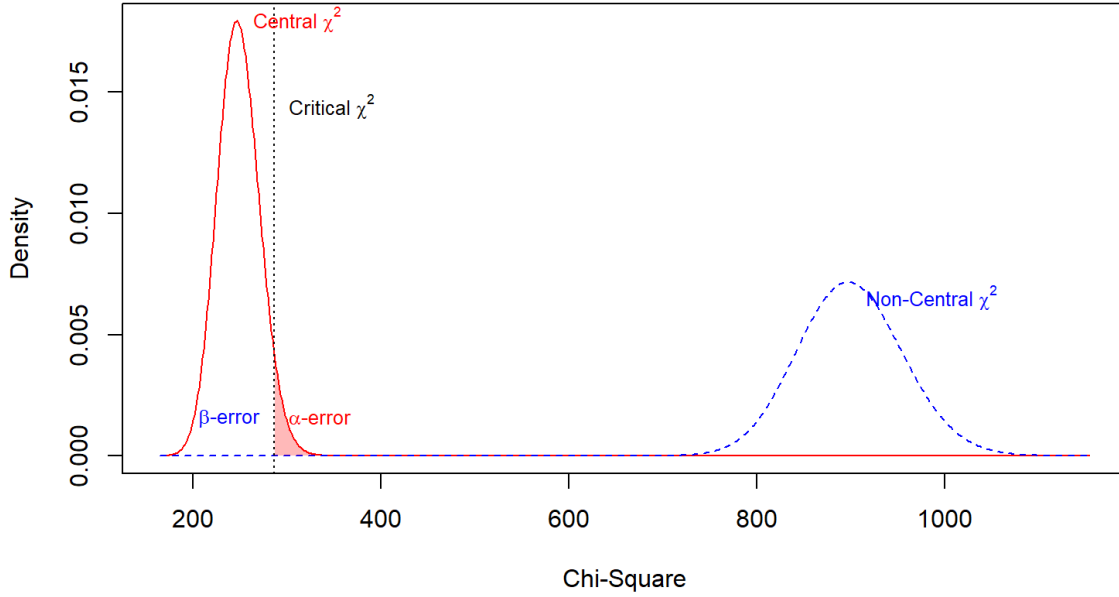


Figure 24: Post hoc power analysis for RMSEA = 0.05.

Source: [Post Hoc Power Analysis](#)

With $F_0 = 0.623$, RMSEA = 0.05, $df = 249$, and $N = 1047$, the noncentrality parameter reached $NCP = 651.135$, far exceeding the critical value $\chi^2_{crit} = 286.808$. Power was effectively 1. The noncentral χ^2 curve lies far to the right of the critical value in the figure, and the shaded β region is negligible (Jobst et al., 2023; MacCallum et al., 1996).

6.2.2 Model-based specification: global fit

The second specification derives the effect from the discrepancy between `popmodel` and the fitted analysis model, evaluating the null model against the saturated model. This tests global fit rather than an isolated restriction, so it is sensitive to any source of misspecification in the analysis model, not only the cross-loadings (Moshagen & Bader, 2024; Wang, 2023).

semPower: Post hoc power analysis

F0	0.303870
RMSEA	0.034934
SRMR	0.175600
Mc	0.859044
GFI	0.975303
AGFI	0.970244
CFI	0.977582
df	249
Num Observations	1047
NCP	317.8482
Critical Chi-Square	286.8077
Alpha	0.050000
Beta	6.740368e-16
Power (1 - Beta)	> 0.9999
Implied Alpha/Beta Ratio	7.417993e+13

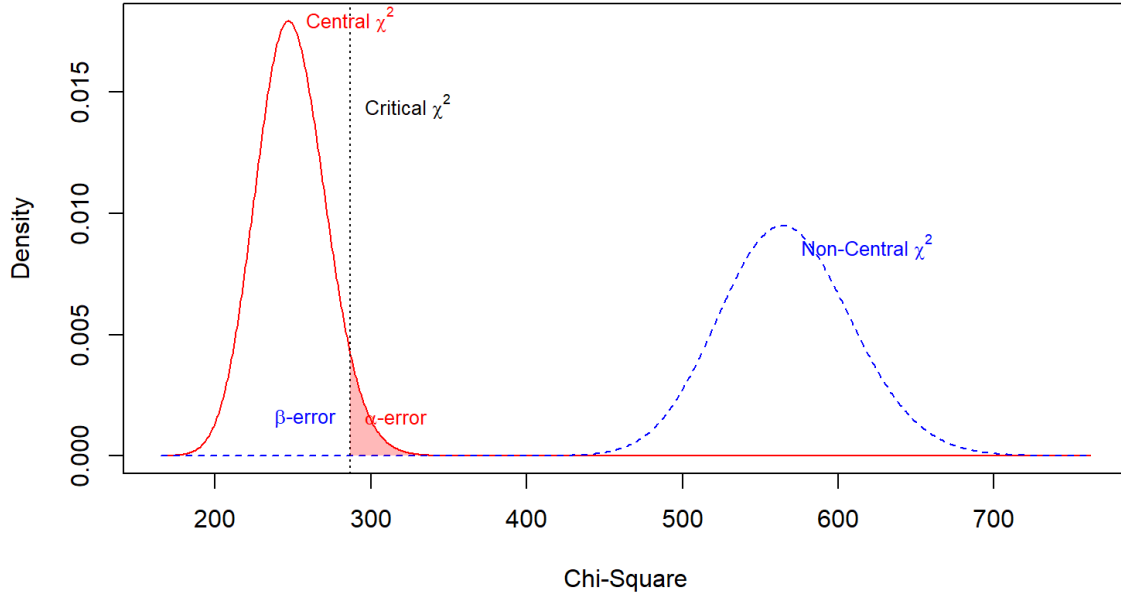


Figure 25: Post hoc power analysis for the real model.

Source: [Post Hoc Power Analysis](#)

With $F_0 = 0.304$, $RMSEA = 0.035$, $df = 249$, and $N = 1047$, the noncentrality reached $NCP = 317.848$ against a critical value of $\chi^2_{crit} = 286.808$. The graphical result is similar to the model-free case — the noncentral distribution lies predominantly to the right of the cutoff — though with slightly less separation, reflecting the more substantively grounded (and generally smaller) effect encoded in the population model.

6.2.3 Model-based specification: nested comparison

The third specification introduces `h1modelfree` as `modelH1`, turning the analysis into a nested comparison between the analysis model and a more permissive alternative that includes the theoretically motivated cross-loadings. Power now refers to the probability of detecting the additional constraints imposed by H_0 relative to this specific competitor — a more targeted and interpretable question than the global test against the saturated model (Jak et al., 2021; Jobst et al., 2023; Moshagen & Bader, 2024).

semPower: Post hoc power analysis

F0	0.009082
RMSEA	0.055021
Mc	0.995469

df	3
Num Observations	1047
NCP	9.499715

Critical Chi-Square	7.814728
Alpha	0.050000
Beta	0.262955
Power (1 - Beta)	0.737045
Implied Alpha/Beta Ratio	0.190147

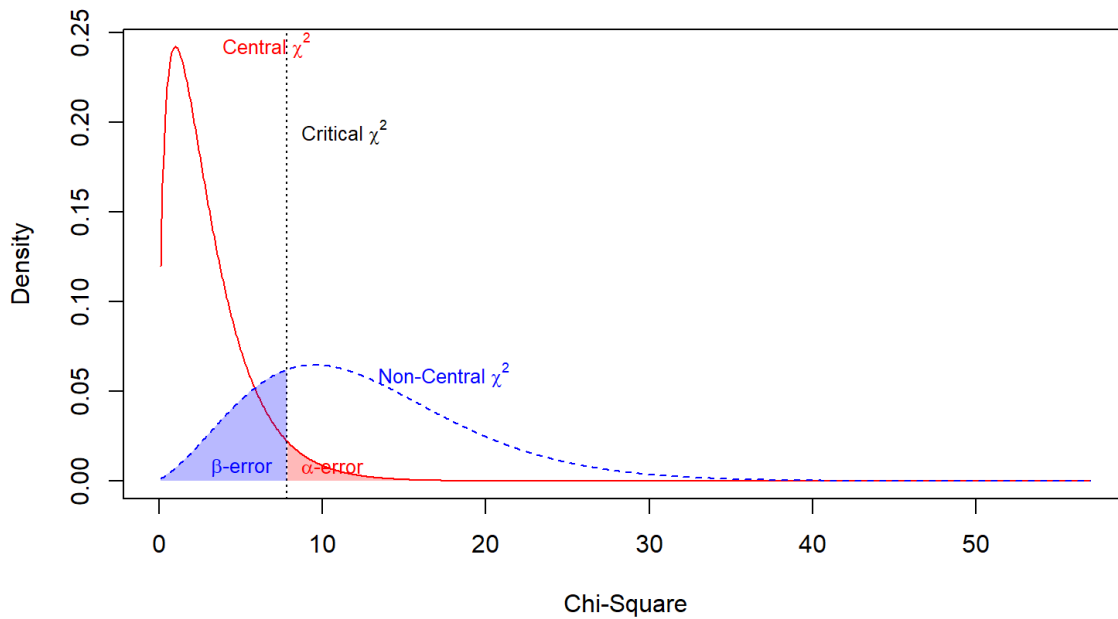


Figure 26: Post hoc power analysis for the misspecified model.

Source: [Post Hoc Power Analysis](#)

The result is considerably more modest. With $F_0 = 0.009$, $RMSEA = 0.055$, and only $df =$

3, the noncentrality reached $NCP = 9.5$. The critical value was $\chi^2_{crit} = 7.815$, and the two distributions overlapped substantially. As a result, $\beta = 0.263$ and power = 0.737, falling below the conventional 0.80 threshold. The noncentral curve is only moderately shifted to the right of the cutoff in the figure, and the shaded β region is visibly large — the empirical counterpart of the insufficient NCP.

This contrast across the three specifications illustrates a fundamental principle: the same N can simultaneously imply very high power for a global test and inadequate power for a more localized nested comparison, because power in SEM is always conditional on the specific discrepancy being tested (Jobst et al., 2023; Moshagen & Bader, 2024). Researchers designing SEM studies should therefore match their power analysis approach to their inferential target — and that choice is not merely statistical, but substantive.

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